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# A New Implementation of Network GARCH Model for Stock Volatility and Co-Volatility Forecasting

Peiyi Zhou 

Department of Statistical Science, University College London, London, UK

Correspondence: Peiyi Zhou ([ucakpz0@ucl.ac.uk](mailto:ucakpz0@ucl.ac.uk))

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## ABSTRACT

Volatility clustering and spillovers are key features of financial time series with many cross-sectional assets. While network analysis links similar or correlated stocks and helps trace volatility spillovers, contemporary multivariate ARCH-GARCH formulations struggle to represent structured network dependence and remain parsimonious. We introduce the generalized network GARCH model under the DCC framework (DCC-GNGARCH), that embeds the GARCH dynamics within the generalized network autoregressive (GNAR) framework to capture an asset's volatility driven by its own history and by neighboring assets in a constructed network. DCC-GNGARCH also extends existing network GARCH formulations by adapting neighboring volatility persistence, dynamic conditional covariance updates, and allowing higher order neighboring effects beyond immediate neighbors. This paper provides the model derivation, vectorization and conversion, and an extension by incorporating threshold coefficients to capture leverage effects. We show that the DCC-GNGARCH is a valid volatility model satisfying the stylized facts of financial return series through simulation. Parameter estimation is then performed by using squared returns as variance proxy and minimizing the negative log-likelihood (NLL) loss function. We apply our model on 75 of the most active US stocks under a constructed network and highlight the model's ability in volatility estimation and forecast, whereas robustness checks under alternative network constructions, different proxy choices, error specifications, and sparsity levels confirm the stability of the main empirical findings.

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## 1 | Introduction

In finance, the term “volatility” is commonly known as the degree of variation of prices disregarding directions, and we usually see the tendency of high- and low-volatility periods to occur in clusters for financial returns. This fundamental phenomenon is named volatility clustering, and related time series models are classified as volatility models.

Classical time series volatility models, most notably the autoregressive conditional heteroskedasticity (ARCH) model (Engle 1982) and the generalized autoregressive conditional heteroskedasticity (GARCH) extension (Bollerslev 1986),

have been adapted for capturing dynamic volatility in both finance and other interdisciplinary studies. Later, Engle and Kroner (1995) and Engle (2002) discussed the extension of univariate ARCH-GARCH to multivariate analysis, in the form of the Baba, Engle, Kraft, and Kroner GARCH (BEKK-GARCH) model and the dynamic conditional correlation GARCH (DCC-GARCH) model, respectively. These studies have greatly enhanced the applicability of ARCH-GARCH models in multiple dimensions, in the context of many cross-sectional financial stock return series. However, real-world applications often involve complex interdependencies among variables. Assets being similar (for example, sharing the same or similar sector affiliation and supply-chain exposure) or

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correlated (showing high correlation in prices/returns) are more likely to influence each other and exhibit stronger volatility spillovers, where shocks propagate normally through constructed network effects rather than isolated pairwise relationships. These contemporary multivariate ARCH-GARCH volatility models mentioned above may not capture such network interactions well, while also suffering severe complexity because many parameters are needed.

Network analysis therefore offers a parsimonious framework for modeling these complex interdependencies. In recent years, with the development of big data and deep learning methods, researchers have increasingly focused on the integration of network and time series analysis. Knight et al. (2016) first introduced the idea of network time series with a specified network autoregressive (integrated) moving average (NARIMA) model, Zhu et al. (2017) then proposed the network vector autoregressive (NAR) model, and Knight et al. (2020) popularized the generalized network autoregressive (GNAR) model, which combines the classic autoregressive (AR) processes with network structures to provide a novel approach for high-dimensional network time series analysis. Subsequently, Nason et al. (2026) applied GNAR to analyze COVID-19 hospitalizations, demonstrating its strength in real-world applications and parsimony in model setting. Nonetheless, due to its linearity, GNAR can hardly be seen as a model to investigate the dynamic volatility.

It is then very natural to embed GARCH within the GNAR framework to implement a network GARCH volatility model to accurately describe both volatilities of asset returns and network dependence. Some existing work, such as the network GARCH model discussed by Zhou et al. (2020), extends GARCH to a reasonable network context, and Pan and Pan (2024) further improved the proposed network GARCH model by incorporating the threshold structure on model parameters to model volatility asymmetry and capture leverage effects, while retaining the network framework. These models, although parsimonious, have important limitations, in that they (i) neglect the neighboring volatility persistence, which includes information on all neighboring nodes' past volatilities; (ii) only focus on conditional variances based on past information, but drop the covariance updates, which is unrealistic for multivariate analysis; and (iii) restrict interactions to first-order neighbors, which are directly connected neighboring nodes, and therefore omit any possible higher order neighboring effects that can affect volatilities along longer network paths. Therefore, this paper aims to implement a new network GARCH volatility model by adapting GNAR concepts, and address the listed limitations of contemporary network GARCH models.

This paper introduces the generalized network GARCH model under DCC framework (DCC-GNGARCH) in Section 2, with the explicit derivation, model vectorization, and investigation into conversion; in accordance with Pan and Pan (2024), we also briefly introduce how to extend our DCC-GNGARCH with threshold coefficients. Then, Section 3 tests our model via simulated data to show that the proposed volatility model is valid for satisfying the stylized facts; meanwhile, we describe the practical parameter fitting schemes by numerical optimization with squared return as the variance proxy, and a loss function that

is either mean squared error (MSE) or negative log-likelihood (NLL). We show that NLL, under normality assumption, is invariant up to an additive constant and an overall scale factor with the well-known quasi-likelihood (QLIKE) function, and the parameter estimates via it are known as the generally preferred quasi-maximum likelihood estimates (QMLE). Finally, Section 4 applies the QMLE-fitted DCC-GNGARCH model to a real-world dataset of the most active US stocks to illustrate its ability in volatility estimation and forecast under a constructed network, along with robustness checks under alternative network constructions, different proxy choices, error specifications, and sparsity levels to examine the stability of our DCC-GNGARCH model across different scenarios.

## 2 | DCC-GNGARCH Model and Methods

Extended from the univariate time series, a network time series can be written as a tuple of a multivariate time series  $\mathbf{X}_t = (X_{1,t}, \dots, X_{d,t})^T$  on a network  $\mathcal{G} = (\mathcal{X}, \mathcal{E})$  with  $\mathcal{X} = \{1, \dots, d\}$  as the set of total  $d$  nodes/vertices and  $\mathcal{E}$  as the set of edges. Each single entry  $X_{i,t} \in \mathbb{R}$  is the value of univariate time series for node  $i$  in the network  $\mathcal{G}$  at time  $t$ . In other words, a node here corresponds to a single time series variable, and edges encode relationships or connections between those variables.

### 2.1 | Related Network Notions and Notations

The formulation of a network time series model is driven by its underlying graph topology. Throughout the rest of this paper, we will only consider the undirected, unweighted networks without self-loops. This simplification is not only analytically trivial but also reasonable in real financial applications, where we do not believe that stocks influence each other asymmetrically and do not have reliable prior information on connection strengths, so every link is simply treated equally. Furthermore, the absence of self-loops is because each stock's own past volatility is already captured by its own GARCH component, allowing such self-edge would then only duplicate that effect and add no new information. Therefore, we can further simplify key network notions:  $r$ -stage neighbors, adjacency matrices, and connection weights introduced by Knight et al. (2016), Knight et al. (2020), and Nason et al. (2026).

**Definition 1.**  $r$ -stage neighbor We say node  $j$  is an  $r$ -stage neighbor of node  $i$  if, in a network  $\mathcal{G} = (\mathcal{X}, \mathcal{E})$ , the shortest path connecting node  $i$  and node  $j$  is of length  $r$ . Following this, we write  $N_r(i)$  as the set of nodes being as the  $r$ -stage neighbors of node  $i$ , and  $|N_r(i)|$  represents the total number of  $r$ -stage neighbors for node  $i$ .

**Definition 2.**  $r$ -stage adjacency matrix We define a matrix  $\mathbf{S}_r$  that

$$(\mathbf{S}_r)_{ij} = \begin{cases} 1 & \text{if node } j \text{ is the } r\text{-stage neighbor of node } i \\ 0 & \text{otherwise} \end{cases}$$

and such matrix  $\mathbf{S}_r$  is known as the  $r$ -stage adjacency matrix.

For an undirected network, we always have the symmetry of  $\mathbf{S}_r$ , examining  $i$ -th row of  $\mathbf{S}_r$  directly yields the set of  $r$ -stage neighbors of node  $i$ . Obviously, when  $r = 1$ , our  $\mathbf{S}_1$  is the commonly used adjacency matrix.

Another important notion describing the network structure would be the connection weights between nodes. Each node connection weight  $w_{ij}$  with  $i, j \in \mathcal{N}$  lies in  $[0, 1]$  quantifying “the relevance node  $j$  has on node  $i$  with respect to neighborhood regression” (Nason et al. 2026). As we consider the unweighted graph, one way to define the node connection weight  $w_{ij}$  would be the reciprocal of the number of  $r$ -stage neighbors of  $i$ ; hence, each  $r$ -stage neighbor of node  $i$  would be treated in equal importance.

**Definition 3.** Connection weight matrix Knight et al. 2020 For a network  $\mathcal{G} = (\mathcal{N}, \mathcal{E})$  with total  $d$  nodes and undirected, unweighted edges, we define the connection weight matrix  $\mathbf{W} \in \mathbb{R}^{d \times d}$  that

$$w_{ij} = \begin{cases} 1/|N_r(i)| & \text{if node } j \text{ is the } r\text{-stage neighbor of node } i \\ 0 & \text{if node } j \text{ is not a neighbor of } i \text{ at any stage} \end{cases}$$

Furthermore, if the network has distance- or similarity-associated edges, let the distance and similarity/weight between node  $i$  and  $j$  be  $d_{ij}$  and  $\mu_{ij}$ , respectively, then the connection weight matrix  $\mathbf{W}$  now is represented as

$$w_{ij} = \begin{cases} \frac{d_{ij}^{-1}}{\sum_{\ell \in N_r(i)} d_{i\ell}^{-1}} & \text{if node } j \text{ is the } r\text{-stage neighbor of node } i, \text{ distance-based} \\ \frac{\mu_{ij}}{\sum_{\ell \in N_r(i)} \mu_{i\ell}} & \text{if node } j \text{ is the } r\text{-stage neighbor of node } i, \text{ similarity-based} \\ 0 & \text{if node } j \text{ is not a neighbor of } i \text{ at any stage} \end{cases}$$

Unlike most weight matrices, the connection weight matrices  $\mathbf{W}$  are usually asymmetric, even our network is assumed to be undirected, unweighted and no self-loops.

## 2.2 | DCC-GNGARCH Model

Drawing on the GNAR framework of Knight et al. (2020) and Nason et al. (2026), we replace the AR component with a univariate GARCH process to define our global generalized network GARCH under the DCC framework (Engle and Sheppard 2001) with zero conditional mean. This is shortened as global DCC( $M, N$ )-GNGARCH( $p, q, [s_1, \dots, s_q], [r_1, \dots, r_p]$ ) defined as

$$\mathbf{X}_t | \mathcal{F}_{t-1} \sim D(0, \Sigma_t) = D(0, \mathbf{D}_t \mathbf{R}_t \mathbf{D}_t) \quad (1a)$$

$$\begin{aligned} \sigma_{i,t}^2 = & \underbrace{\alpha_0 + \sum_{k=1}^q \alpha_k X_{i,t-k}^2 + \sum_{\ell=1}^p \gamma_{\ell} \sigma_{i,t-\ell}^2}_{\text{own GARCH}(p,q)\text{component}} \\ & + \left[ \underbrace{\sum_{k=1}^q \sum_{r=1}^{s_k} \beta_{kr} \sum_{j \in N_r(i)} w_{ij} X_{j,t-k}^2}_{\text{neighboring clustering effect}} + \underbrace{\sum_{\ell=1}^p \sum_{r'=1}^{r_{\ell}} \delta_{\ell r'} \sum_{j \in N_{r'}(i)} w_{ij} \sigma_{j,t-\ell}^2}_{\text{neighboring persistence effect}} \right] \end{aligned} \quad (1b)$$

$$\sigma_{ij,t} = \sigma_{i,t} \sigma_{j,t} \rho_{ij,t}, \quad \rho_{ij,t} = (\mathbf{R}_t)_{ij} \quad (1c)$$

where  $D$  in (1a) means a certain type of distribution, and we normally set it as Gaussian distribution for Gaussian noise. Our setting of  $\mathbf{D}_t$  and  $\mathbf{R}_t$  follows Engle and Sheppard (2001). We define  $\mathbf{D}_t$  in (1a) as the diagonal matrix of conditional standard deviations

$$\mathbf{D}_t := \text{diag}(\sqrt{\sigma_{1,t}^2}, \dots, \sqrt{\sigma_{d,t}^2}) \quad (2)$$

Let the standardized residuals be  $\epsilon_t := \mathbf{D}_t^{-1} \mathbf{X}_t$  and  $\bar{\mathbf{Q}}$  denote the unconditional covariance matrix of  $\epsilon_t$  (estimated from the first-stage standardized residuals), which is  $\bar{\mathbf{Q}} = \text{cov}(\epsilon_t) = \mathbb{E}(\epsilon_t \epsilon_t^T)$  being positive definite and commonly estimated as  $\hat{\bar{\mathbf{Q}}} = \frac{1}{T} \sum_{t=1}^T \epsilon_t \epsilon_t^T$  (Nordström 2021). Define for  $t \in \mathbb{N}$  that

$$\mathbf{Q}_t = \left( 1 - \sum_{m=1}^M \lambda_m - \sum_{n=1}^N \omega_n \right) \bar{\mathbf{Q}} + \sum_{m=1}^M \lambda_m (\epsilon_{t-m} \epsilon_{t-m}^T) + \sum_{n=1}^N \omega_n \mathbf{Q}_{t-n} \quad (3)$$

and  $\mathbf{R}_t$  being a time-varying symmetric conditional correlation matrix modeled by a DCC( $M, N$ ) framework is obtained by normalization that, with  $\mathbf{Q}_t^* = \text{diag}(\mathbf{Q}_t)^{1/2} = \text{diag}(q_{11,t}^{1/2}, \dots, q_{dd,t}^{1/2})$ ,

$$\mathbf{R}_t = (\mathbf{Q}_t^*)^{-1} \mathbf{Q}_t (\mathbf{Q}_t^*)^{-1} \Leftrightarrow \rho_{ij,t} = \frac{q_{ij,t}}{\sqrt{q_{ii,t} q_{jj,t}}} \quad (4)$$

“Global” means that we treat with global parameters

$$\left[ \alpha_0, \{\alpha_k\}_{k=1}^q, \{\gamma_{\ell}\}_{\ell=1}^p, \{\beta_{kr}\}_{(k=1, r=1)}^{(q, s_k)}, \{\delta_{\ell r'}\}_{(\ell=1, r'=1)}^{(p, r_{\ell})} \right] \quad (5)$$

across the whole network. Although allowing local parameters depending on the specific nodes will capture individual node behavior more accurately, that extension substantially increases the model complexity. Therefore, we focus here on the global model only, and all DCC-GNGARCH models in this paper use this global specification. The local parameter analysis is deferred to Supporting Information S1.

It is also important to clarify the role of the network structure in the DCC component. The network matrices  $\mathbf{W}$  and  $\mathbf{S}_r$  enter the model explicitly through the conditional variance recursion (1b). By contrast, the DCC correlation recursion in (3) and (4) is not directly network-augmented, since no network-based term appears in the update of  $\mathbf{Q}_t$  and  $\mathbf{R}_t$ . However, the network still affects the DCC component indirectly through the standardized residuals  $\epsilon_t$ , as  $\mathbf{D}_t$  is determined by the network GARCH variance recursion. Thus, in the present specification, network effects are modeled explicitly in the volatility dynamics, whereas a directly network-augmented correlation recursion can be a possible extension.

Through the above equations,  $\mathbf{X}_t = (X_{1,t}, \dots, X_{d,t})^T$  and  $X_{i,t}$  denotes the time series variable observed at node  $i$  and time  $t$ , assumed to have zero conditional mean at all time. In finance, we usually treat  $X_{i,t}$  as the return (either “mean zero” or the “residuals” from a filtered time series Engle and Sheppard 2001) of asset  $i$  at time  $t$  once each node represents a financial asset, either in the form of simple return or log-return.

**Definition 4.** Financial returns If  $\{P_t\}_{t=0}^T$  is a sequence of observed prices for a stock, then for  $t = 1, \dots, T$ , its simple return and log-return are defined, respectively, as

$$\text{simple return: } r_t = \frac{P_t - P_{t-1}}{P_{t-1}} \quad (6a)$$

$$\text{log-return: } r_t = \log P_t - \log P_{t-1} \quad (6b)$$

$\sigma_{i,t}^2$  is conditional variance of  $X_{i,t}$  and  $\sigma_{ij,t}$  is the conditional covariance between  $X_{i,t}$  and  $X_{j,t}$  given all past information  $\mathcal{F}_{t-1}$ , where  $\mathcal{F}_{t-1} = \bigcup_{i=1}^d \mathcal{F}_{i,t-1}$  with every  $\mathcal{F}_{i,t-1} = \sigma(X_{i,t-1}, \dots, X_{i,0})$ .  $\Sigma_t$  is the overall conditional covariance matrix with  $(\Sigma_t)_{ii} = \sigma_{i,t}^2$  and  $(\Sigma_t)_{ij} = \sigma_{ij,t}$ . Lastly,  $\mathbf{Z}_t = (Z_{1,t}, \dots, Z_{d,t})^T$  is a vectorized strict white noise (SWN) variable with zero mean  $\mathbb{E}[\mathbf{Z}_t] = 0$  and unit variance  $\text{var}(\mathbf{Z}_t) = \mathbf{I}_d$ , where every vectorized variable in the SWN process  $\{\mathbf{Z}_t\}$  is independent and identically distributed (i.i.d.) across both time  $t$  and assets  $i$ .

As asset volatility is generally known as a latent and unobservable variable, we should apply proxies to invigilate volatility. If we already fit a volatility model, then it is common to use the conditional standard deviation  $\sigma_{i,t}$  for volatility as a more direct measure. See more about volatility/variance proxy analysis in Section 3.3.1.

### 2.2.1 | Model Likelihood Estimation

As our model is also under the DCC framework, its likelihood should intuitively in a similar form as the classic DCC-GARCH likelihood as shown in Engle and Sheppard (2001). Following our setting of model, if we further assume the normality assumption, the full density among all historical values of  $\{\mathbf{X}_t\}_{t=1}^T$  can be expressed as

$$p(\mathbf{X}_1, \dots, \mathbf{X}_T) = \prod_{t=1}^T p(\mathbf{X}_t | \mathcal{F}_{t-1}) = \prod_{t=1}^T \mathcal{N}(\mathbf{X}_t; 0, \mathbf{D}_t \mathbf{R}_t \mathbf{D}_t)$$

so then the log-likelihood (and it is indeed quasi log-likelihood under Gaussianity) is written as

$$\begin{aligned} L(\theta, \phi) &= \sum_{t=1}^T \log \mathcal{N}(\mathbf{X}_t; 0, \mathbf{D}_t \mathbf{R}_t \mathbf{D}_t) \\ &= -\frac{1}{2} \sum_{t=1}^T (d \log(2\pi) + \log |\mathbf{D}_t \mathbf{R}_t \mathbf{D}_t| + \mathbf{X}_t^T (\mathbf{D}_t \mathbf{R}_t \mathbf{D}_t)^{-1} \mathbf{X}_t) \\ &= -\frac{1}{2} \sum_{t=1}^T (d \log(2\pi) + 2 \log |\mathbf{D}_t| + \log |\mathbf{R}_t| + \epsilon_t^T \mathbf{R}_t^{-1} \epsilon_t) \end{aligned} \quad (7)$$

where as stated above that  $\epsilon_t := \mathbf{D}_t^{-1} \mathbf{X}_t$  as the standardized residuals,  $\theta$  represents all parameters related to (1b), and  $\phi$  is for parameters associated with (1c). We then apply the same technique as Engle (2002) that rewrite the joint log-likelihood (7) as the sum of two terms: volatility part  $L_v$  and correlation part  $L_c$ , that

$$L(\theta, \phi) = L_v(\theta) + L_c(\theta, \phi),$$

$$\begin{cases} L_v(\theta) = -\frac{1}{2} \sum_{t=1}^T \sum_{i=1}^d \left( \log(2\pi) + \log(\sigma_{i,t}^2) + \frac{X_{i,t}^2}{\sigma_{i,t}^2} \right) \\ L_c(\theta, \phi) = -\frac{1}{2} \sum_{t=1}^T (\log |\mathbf{R}_t| + \epsilon_t^T \mathbf{R}_t^{-1} \epsilon_t - \epsilon_t^T \epsilon_t) \end{cases}$$

so we can apply two-step QMLE by first maximizing  $L_v(\theta)$ , and then inserting  $\theta^*$  that maximizes  $L_v(\theta)$  to find the optimal  $\phi^*$  that maximizes  $L_c(\theta^*, \phi)$ .

Although we have assumed Gaussianity for the target variable  $\mathbf{X}_t | \mathcal{F}_{t-1}$ , Bollerslev and Wooldridge (1992) discussed that we only need that assumption to construct our log-likelihood function, whereas in volatility models, it is not necessary to assume that  $\mathbf{X}_t | \mathcal{F}_{t-1}$  follows a conditional normal distribution. The resulting QMLE is generally consistent for the parameters analyzed:  $(\theta, \phi)$ , and satisfies an asymptotic normality result once standard regularity requirements are met. For DCC-related models, such asymptotic normality should be in the same form as Engle and Sheppard (2001).

### 2.2.2 | Positive Definiteness of Model Conditional Covariance

To ensure our setting of our DCC( $M, N$ )-GNGARCH ( $p, q, [s_1, \dots, s_q], [r_1, \dots, r_p]$ ) model is statistically sensible, it needs to assure: variance-covariance consistency  $\sigma_{i,t}^2 = \sigma_{ii,t}$ ; covariance symmetry  $\sigma_{ij,t} = \sigma_{ji,t}$  and positive definiteness of overall conditional covariance matrix  $\Sigma_t$ . The former two are obviously true from our definition of symmetric  $\mathbf{R}_t$  (4), and for the last one, as our model is DCC-based we shall extend Proposition 2 in Engle and Sheppard (2001).

**Proposition 1.** Positive definiteness of model conditional covariance With  $\Sigma_t = \mathbf{D}_t \mathbf{R}_t \mathbf{D}_t$  as our DCC( $M, N$ )-GNGARCH ( $p, q, [s_1, \dots, s_q], [r_1, \dots, r_p]$ ) model conditional covariance matrix as defined in (1a), with  $\mathbf{D}_t$  and  $\mathbf{R}_t$  defined as (2) and (4), respectively, if we have the following assumptions:

- $\alpha_0 > 0$  and proper  $\alpha_k, \gamma_\ell, \beta_{kr}, \delta_{\ell r'}$  for all  $k \in \{1, \dots, q\}$ ,  $\ell \in \{1, \dots, p\}$ ,  $r = s_1, \dots, s_q$  and  $r' = r_1, \dots, r_p$  that ensures  $\sigma_{i,t}^2 > 0$  with probability one and (weak) stationary for all  $i, t$ .
- DCC parameters satisfying  $\lambda_m \geq 0, \omega_n \geq 0, \sum_{m=1}^M \lambda_m + \sum_{n=1}^N \omega_n < 1$ .
- $\mathbf{Q}_0 > 0$  is bounded away from the singularity (i.e.  $\lambda_{\min}(\mathbf{Q}_0) \geq \delta > 0$ ), for numerical stability.

Then  $\Sigma_t$  is positive definite for all  $t$ .

*Proof.* The idea of the proof is same as the proof of Proposition 2 in Engle and Sheppard (2001) as we merely replace the original setting for the univariate GARCH conditional variance components into the DCC-GNGARCH conditional variance



components as (1b). Explicitly, we show  $\mathbf{Q}_t$  as (3) is positive definite for all  $t$ , since it is a weighted average of a positive definite matrix ( $\bar{\mathbf{Q}}$ ), positive semi-definite matrices ( $\epsilon_{t-m}\epsilon_{t-m}^T$ ) and a positive definite matrices  $\mathbf{Q}_{t-1}$ , with  $\mathbf{Q}_0$  being positive definite by Assumption (iii), and then by Proposition 1 of Engle and Sheppard (2001) to state  $\mathbf{R}_t$  is positive definite.  $\square$

In practice,  $\mathbf{Q}_0$  is normally initialized as a positive definite estimate of  $\bar{\mathbf{Q}}$  to avoid near-singularity in finite samples (so Assumption (iii) nearly always works). Consequently, beyond the DCC parameter constraints, it remains to establish conditions ensuring the existence stationarity of the network GARCH variance recursion (1b) (Assumption (i) in Proposition 1). Similar to the univariate GARCH model, in order to ensure the positivity of conditional variance in (1b), a common and sufficient condition on these global parameters is

$$\alpha_k \geq 0, \gamma_\ell \geq 0, \beta_{kr} \geq 0, \delta_{\ell r'} \geq 0 \quad (8)$$

for  $k = 1, \dots, q$ ,  $\ell = 1, \dots, p$ ,  $r = s_1, \dots, s_q$  and  $r' = r_1, \dots, r_p$ .

**Theorem 1.** Sufficient condition for stationarity of (1b), static network. If global parameters  $\alpha_k, \gamma_\ell, \beta_{kr}, \delta_{\ell r'}$  for all  $k \in \{1, \dots, q\}, \ell \in \{1, \dots, p\}, r = s_1, \dots, s_q$  and  $r' = r_1, \dots, r_p$  hold

$$\sum_{k=1}^q \left( |\alpha_k| + \sum_{r=1}^{s_k} |\beta_{kr}| \right) + \sum_{\ell=1}^p \left( |\gamma_\ell| + \sum_{r'=1}^{r_\ell} |\delta_{\ell r'}| \right) < 1$$

together with the non-negativity constraints (8), then (1b) is stationary for all  $i$  and  $t$ .

*Proof.* The proof requires us to vectorize (1b) and derive an associated VARMA representation for the squared process  $\mathbf{X}_t^2$ , which enables a simple sufficient stability condition paralleling the GNAR-VAR analysis of Knight et al. (2020). Note that such VARMA representation is not required for empirical estimation, but it provides a transparent theoretical justification for stationarity of the variance dynamics in the proposed network setting. For details, see Supporting Information S2.  $\square$

### 2.2.3 | Finiteness of the Unconditional Expectation of Conditional Covariance

Building on the positive definiteness established in Proposition 1 and the variance stability condition in Theorem 1, we now establish the boundedness of the unconditional expectation of conditional covariance matrix for the DCC-GNGARCH process.

**Proposition 2.** Finiteness of unconditional expectation of conditional covariance. Assume the variance process  $\{\mathbf{h}_t\}$  that  $\mathbf{h}_t = (\sigma_{1,t}^2, \dots, \sigma_{d,t}^2)^T \in \mathbb{R}^d$  satisfies the sufficient stationarity conditions of Theorem 1, implying that  $\{\mathbf{h}_t\}$  is weakly stationary and the unconditional variances are finite ( $\mathbb{E}(\sigma_{i,t}^2) = \bar{h}_i < \infty$ ). Suppose further that the DCC parameters satisfy the positive definiteness and persistence constraints of Proposition 1. Then, every element of the conditional covariance matrix  $\Sigma_t = \mathbf{D}_t \mathbf{R}_t \mathbf{D}_t$

has a finite unconditional expectation, meaning  $\mathbb{E}(\Sigma_t)$  is bounded and well-defined.

*Proof.* By Proposition 1,  $\mathbf{R}_t$  is a well-defined correlation matrix for all  $t$ . Therefore, its off-diagonal elements are strictly bounded such that  $|\rho_{ij,t}| \leq 1$ . The elements of the conditional covariance matrix  $\Sigma_t$  are given by  $\sigma_{ij,t} = \sigma_{i,t} \sigma_{j,t} \rho_{ij,t}$ . It follows that the absolute magnitude of the conditional covariance is bounded by the product of the conditional standard deviations

$$|\sigma_{ij,t}| \leq \sigma_{i,t} \sigma_{j,t}$$

Taking the unconditional expectation and applying the Cauchy–Schwarz inequality yields

$$\mathbb{E}(|\sigma_{ij,t}|) \leq \mathbb{E}(\sigma_{i,t} \sigma_{j,t}) \leq \sqrt{\mathbb{E}(\sigma_{i,t}^2) \mathbb{E}(\sigma_{j,t}^2)} = \sqrt{\bar{h}_i \bar{h}_j} < \infty$$

as under the conditions of Theorem 1, the unconditional variances are finite. Because all elements of the matrix are bounded by finite constants, the unconditional expectation of the overall covariance matrix,  $\mathbb{E}(\Sigma_t)$ , is finite.  $\square$

### 2.2.4 | Model Overall Stationarity

The overall weak stationarity of the proposed DCC-GNGARCH model is, however, more delicate to establish. By Tsay (2010a), a multivariate time series is weakly stationary if its first and second moments are time-invariant and its cross-covariance matrix depends only on the time lag. In our setting, it is straightforward to verify that  $\mathbb{E}(\mathbf{X}_t) = 0$  and  $\text{cov}(\mathbf{X}_t, \mathbf{X}_s) = 0$  for  $t \neq s$  by conditioning on  $\mathcal{F}_{t-1}$  and applying the tower property. Moreover, Proposition 2 guarantees that the unconditional second moments of the system exist and are finite. The main difficulty lies in proving that  $\mathbb{E}(\Sigma_t)$  is exactly time-invariant. Indeed, under the DCC decomposition  $\Sigma_t = \mathbf{D}_t \mathbf{R}_t \mathbf{D}_t$ , the nonlinear normalization  $\mathbf{R}_t = (\mathbf{Q}_t^*)^{-1} \mathbf{Q}_t (\mathbf{Q}_t^*)^{-1}$  couples the variance and correlation dynamics in a way that prevents a simple analytical characterization of  $\mathbb{E}(\Sigma_t)$ .

This difficulty is already well known in the DCC literature. For the standard DCC-GARCH model, Fermaian and Malongo (2017) derives sufficient conditions for the existence and uniqueness of a strictly (and also weakly) stationary solution by representing the model as a nonlinear Markov chain and applying Tweedie (1988) method. Their analysis shows that, for DCC-type models, stationarity is determined by the dynamics of the full latent state process rather than by a simple persistence inequality like univariate GARCH. This is reflected in the fact that the DCC correlation recursion is not autonomous, that the recursion is written for  $\mathbf{Q}_t$  (3), its driving term involves the standardized residuals, and these depend on the volatility state through  $\mathbf{D}_t$  (2); the conditional correlation matrix  $\mathbf{R}_t$  is then obtained from  $\mathbf{Q}_t$  through the nonlinear normalization (4). As a result, stationarity must be studied for the joint latent state process rather than for the correlation recursion in isolation.

Meanwhile, in the corrected DCC (cDCC) framework, Aielli (2013) gives a stationarity principle in Proposition 2.3, that under suitable assumptions on the correlation recursion, the process formed by the correlation and the corresponding standardized innovations admits a non-anticipative, strictly and weakly stationary, and ergodic solution. With an additional measurability condition on the variance equations, the full state process is strictly stationary and ergodic; with finite second moments, the cDCC return process is weakly stationary.

For the present DCC-GNGARCH specification, these references are informative rather than directly applicable. Our model preserves the same DCC correlation structure, but replaces the usual univariate GARCH variance block by a network GARCH recursion. It is therefore natural to expect that an analogous strict stationarity argument could be developed for the joint network-augmented state process, but doing so would require extending the nonlinear Markov chain analysis of the DCC literature to the present setting, which is beyond the scope of this paper. Consequently, we confine ourselves here to explicit sufficient conditions ensuring positivity, finite second moments, and empirical stability of both the variance and correlation components, while leaving a full proof of strict stationarity and ergodicity for the DCC-GNGARCH system to future theoretical work.

### 2.2.5 | Network Volatility Autocorrelation Function

Finally, inspired by the network autocorrelation function (NACF) introduced by Nason et al. (2026), which defines in GNAR for graphical aids and quantifies the correlation observed in the network with respect to the  $l$ -lag,  $r$ -stage neighbors across all nodes, we can follow this idea to define network volatility autocorrelation function (NVACF) at pair  $(l, r)$  for our volatility model DCC-GNGARCH.

**Definition 5.** Network volatility autocorrelation function (NVACF) With connection weight matrix  $\mathbf{W}$  and  $r$ -stage adjacency matrix  $\mathbf{S}_r$ , for a network with total  $d$  nodes, the NVACF of a volatility model DCC-GNGARCH is defined in terms of

$$\text{nvacf}(l, r) = \frac{\sum_{t=1}^{T-l} (\mathbf{h}_{t+l} - \bar{\mathbf{h}})^T (\mathbf{W} \odot \mathbf{S}_r + \mathbf{I}_d) (\mathbf{h}_t - \bar{\mathbf{h}})}{\sum_{t=1}^T (\mathbf{h}_t - \bar{\mathbf{h}})^T ((1 + \lambda) \mathbf{I}_d) (\mathbf{h}_t - \bar{\mathbf{h}})} \quad (9)$$

where again  $\mathbf{h}_t^2 = (\sigma_{1,t}^2, \dots, \sigma_{d,t}^2)^T \in \mathbb{R}^d$  is the vector of conditional variance for all  $d$  nodes at time  $t$  by (1b), and  $\bar{\mathbf{h}} \in \mathbb{R}^d$  is the corresponding vector of sample means (one mean per node) computed across the time dimension, i.e.,  $(\bar{\mathbf{h}})_i = \sum_{t=1}^T \sigma_{i,t}^2 / T$ , with

$$\text{autocovariance bound: } \lambda = \left[ \max_{j=1, \dots, d} \left\{ \sum_{i=1}^d [\mathbf{W} \odot \mathbf{W}]_{ij} \right\} \right]^{1/2}.$$

Analogously to the NACF, our NVACF harnesses the topology of the network, combining the information from each node and its neighbors in stages  $r$  to calculate the autocorrelation in lag  $l$ . Meanwhile, using the autocovariance bound  $\lambda$  defined in a similar form as NACF in Nason et al. (2026) also ensures that

NVACF values are bounded between  $-1$  and  $1$ , along with  $l$  and  $r$ . On the other hand, NVACF acts as an inferential, model-based plug-in statistic rather than a descriptive one, since volatility is unobservable, and we use the model-implied conditional variance as a proxy to study this latent variable. This makes the NVACF a model-dependent measure that evaluates the network volatility autocorrelation structure implied by the DCC-GNGARCH specification.

To analyze network autocorrelation decay in a multivariate return series, we first fit the data into the DCC-GNGARCH model and then plot the NVACF values via the correlation-orbit (Corbit) plot. This is illustrated later through a simulation study, with the corresponding Corbit plot given in Figure 6.

### 2.3 | Model Extension: DCC-GTN-GARCH

If we observe or believe, as in many real-life cases, the leverage effect, which describes how an asset's volatility tends to move inversely with its returns, that rising prices usually coincide with lower volatility, and vice versa (Ait-Sahalia et al. 2013), then it is useful to further enhance our DCC-GNGARCH model in Section 2.2 by using volatility asymmetry.

Inspired by univariate threshold GARCH (TGARCH) models (Francq and Zakoian 2019) and the recent threshold network GARCH formulation of Pan and Pan (2024), we call our threshold-augmented framework the generalized threshold network GARCH model under the DCC framework (DCC-GTN-GNGARCH) by incorporating threshold on the node's squared return part. Concretely, one obtains DCC( $M, N$ )-GTN-GARCH( $p, q, [s_1, \dots, s_q], [r_1, \dots, r_p]$ ) by replacing each  $\alpha_k$  in (1b) with  $\alpha'_k$  that

$$\alpha'_k = \alpha_k^{(+)} \mathbf{1}(X_{i,t-k} \geq 0) + \alpha_k^{(-)} \mathbf{1}(X_{i,t-k} < 0) \quad (10)$$

The financial intuition behind the setting of  $\alpha_k^{(+)}$ ,  $\alpha_k^{(-)}$  is to explain the leverage effect of financial time series, DCC-GTN-GARCH is definitely a more complicated model, but since we do not change the neighboring effect terms, its statistical properties and vectorization are similar to those of DCC-GNGARCH, and we will give its vectorization in Supporting Information S1.

### 3 | Model Specification, Simulation, and Estimation

As Bollerslev (1986) pointed out, GARCH(1, 1) model can already handle most financial return series without introducing too many parameters from higher order GARCH, see also Tsay (2010b). We choose to concentrate on our parsimonious network-augmented GARCH(1, 1) specification under DCC(1, 1) frame, i.e., DCC(1, 1)-GNGARCH(1, 1, [1], [1]), in which each node's conditional variance depends on its own one-period lag and the one-period lag of its first-stage neighbors. With the notation in (1a-1c), our parameters are  $(\alpha_0, \alpha_1, \gamma_1, \beta_{11}, \delta_{11}, \lambda, \omega)$ , which contains significantly fewer parameters than a classic VARMA(1, 1) model, see Table 1 below.

**TABLE 1** | Comparison of model parameters between DCC-GNGARCH and VARMA.

| Model (of $d$ nodes/dimension $d$ ) | Active parameters |
|-------------------------------------|-------------------|
| DCC(1,1)-GNGARCH(1, 1, [1], [1])    | 7                 |
| VARMA(1,1), no intercept            | $2d^2$            |
| VARMA(1,1), with intercept          | $2d^2 + d$        |

### 3.1 | Model Simulation and Validity

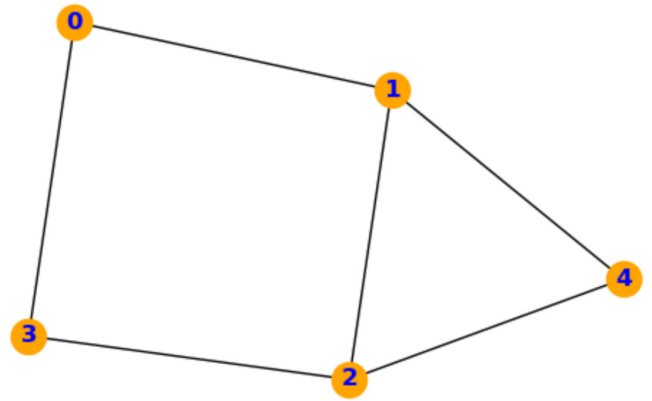
It is always advisable to analyze the proposed model from the simulated data, where we can then demonstrate the validity of the model by examining the stylized facts of the asset returns, which refers to a range of statistical properties of return series consistently observed across diverse instruments, markets, and time periods (Cont 2001). In this paper, we will investigate the following stylized facts:

- SF1. The unconditional distribution of returns has a heavy tail and mild asymmetry.
- SF2. Autocorrelations of asset returns are often negligible that decay to zero rapidly.
- SF3. Volatility clustering and persistence effect: Different measures of volatility shows a positive autocorrelation and slow decay over time lags.
- SF4. Aggregational Gaussianity: The distribution of aggregated returns converges toward a normal distribution as the aggregation timescale increases.

With the same notation as in Section 2.2, we simulate the process  $\{\mathbf{X}_t\}$  under the DCC-GNGARCH model defined by (1a–1c) on the specific five-node undirected, unweighted network shown in Figure 1, under the assumption that the SWN process  $\{\mathbf{Z}_t\}$  is Gaussian. Given initial values  $\mathbf{X}_0$ ,  $\mathbf{D}_0$ ,  $\mathbf{Q}_0$ , and the long-run DCC target matrix  $\bar{\mathbf{Q}}$ , the simulation proceeds recursively for  $t = 1, 2, \dots$ . At each step, we first update the node-wise conditional variances  $\sigma_{i,t}^2$  according to (1b), and hence obtain the diagonal volatility matrix  $\mathbf{D}_t$ . Next, using the standardized residuals from the previous step, we update the DCC matrix  $\mathbf{Q}_t$  through (3), and then normalize it according to (4) to obtain the conditional correlation matrix  $\mathbf{R}_t$ . The conditional covariance matrix is then constructed as  $\Sigma_t = \mathbf{D}_t \mathbf{R}_t \mathbf{D}_t$ , after which  $\mathbf{X}_t$  is generated from the Gaussian distribution  $\mathcal{N}(0, \Sigma_t)$ . In our simulation setting, since the unconditional covariance matrix of the standardized residuals is not available a priori, we take  $\bar{\mathbf{Q}} = \mathbf{I}_d$  as a neutral benchmark, whereas  $\mathbf{Q}_0$  is chosen to be positive definite so the initial correlation matrix  $\mathbf{R}_0$  is well-defined.

We generate a total of 2000 samples of  $\mathbf{X}_t$  through (1a) and discard the first 20% samples (so 400 burnin samples). The “true” parameters we fit for our DCC(1,1)-GNGARCH(1, 1, [1], [1]) model for simulation of  $\{\mathbf{X}_t\}$  are

$$\begin{aligned} \alpha_0 &= 0.05, \alpha_1 = 0.20, \gamma_1 = 0.60, \\ \beta_{11} &= 0.05, \delta_{11} = 0.05, \lambda = 0.4, \omega = 0.3 \end{aligned} \quad (11)$$



**FIGURE 1** | The simple network used for our simulation task and model evaluation. Note that the node index starts from 0 following the Python indexing; hence, here, we have  $\mathbf{X}_t = (X_{0,t}, X_{1,t}, X_{2,t}, X_{3,t}, X_{4,t})^T$ .

For brevity, we present detailed results for node 0, i.e.,  $\{X_{0,t}\}$ , and the remaining nodes exhibit a similar behavior. We begin with the empirical distribution of the simulated returns. Figure 2 overlays the return histogram with both a kernel density estimate (KDE) and the Gaussian density fitted using the sample mean and variance. Pearson's kurtosis and simulated series skewness are computed that

$$\kappa = 3.840 > 3, \quad \beta = 0.174 \neq 0$$

A Pearson kurtosis greater than 3 and a nonzero skewness together indicate the heavy tail and asymmetric nature of simulated return's distribution, aligning with SF1.

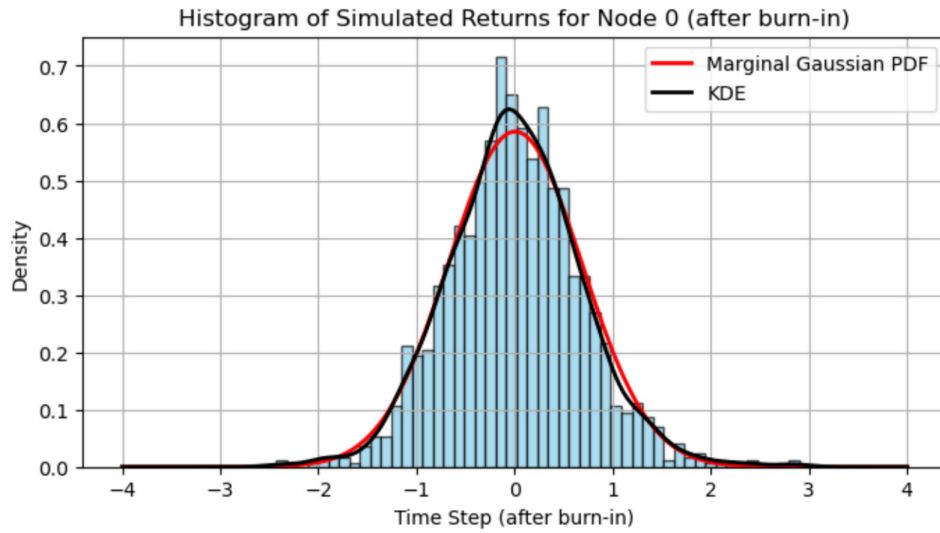
To assess SF2, we compute the sample/empirical autocorrelation function (ACF) of the raw return series

$$\hat{\rho}_k = \frac{\sum_{t=k+1}^T (X_t - \bar{X})(X_{t-k} - \bar{X})}{\sum_{t=1}^T (X_t - \bar{X})^2}$$

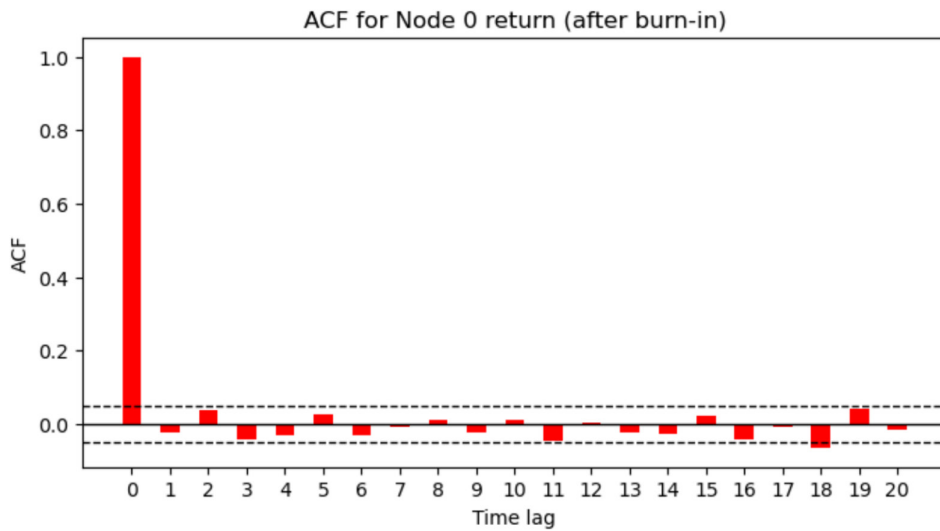
for lags  $k = 1, \dots, 20$ , with the 95% confidence bound of  $\pm 1.96/\sqrt{T}$  (and  $T = 1600$ ). We expect to see most  $\hat{\rho}_k$  fall within the bound, with a rapid decay of ACF. Figure 3 exactly explains the desired trend for the first 20 lags of simulated node 0 returns.

We can assess volatility clustering (SF3) either by computing the sample autocorrelation of the absolute return series  $|X_{i,t}|$  or by using the sequence of model-implied conditional standard deviations  $\{\sigma_{i,t}\}$  directly. Both measures can be seen as our volatility proxies and calculate their ACF, where the positive ACF values and a slow decay with most of them lying outside the 95% confidence bound in Figure 4 demonstrate both volatility clustering and persistence throughout our DCC-GNGARCH volatility model.

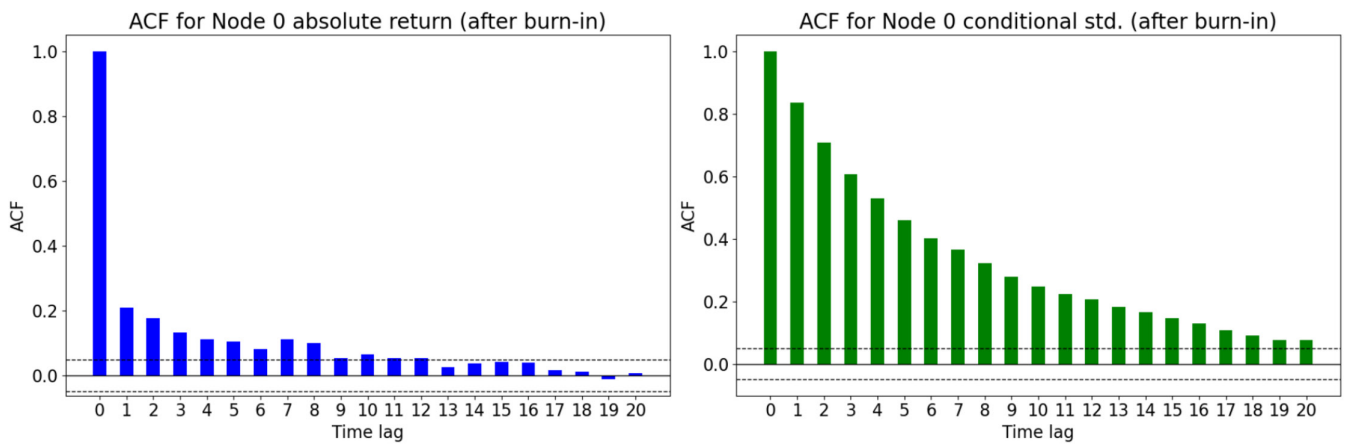
Examine aggregational Gaussianity (SF4) hinges on having enough observations, which depend on the size of the return series. In our case with a return series of length 1600, it is good to set the aggregation timescale to at most 30 days to balance reasonable sample size and aggregation effects. Following this, we evaluate by considering 1- (daily), 7- (weekly), and 30-day (monthly) aggregation windows, and for each, we standardize aggregated returns and plot QQ-plots against the standard



**FIGURE 2** | Histogram of node 0 returns with KDE and normal distribution curve.



**FIGURE 3** | Empirical autocorrelation of the first 20 days (first 20 values) of node 0 return.



**FIGURE 4** | Empirical autocorrelation of the first 20 days of node 0 absolute return (left) and the first 20 lags of simulated conditional standard deviation (right).

normal distribution  $\mathcal{N}(0,1)$ , as shown in Figure 5. The figure indicates that, as the size of aggregation window increases, the empirical distribution becomes more closely aligned with the

Gaussian benchmark, especially in the central quantiles. In particular, the 7-day aggregated returns display a clearer linear pattern than the 1-day returns, whose tails exhibit more



pronounced deviations from the reference line. The 30-day aggregated returns remain broadly consistent with this tendency, although the substantially reduced sample size at that horizon makes the tail behavior more difficult to assess precisely. Overall, the figure provides qualitative evidence in support of aggregational Gaussianity.

### 3.2 | NVACF Illustration With Corbit Plot

The evaluation of NVACF on simulated data from Definition 5 by Corbit plot is shown in Figure 6. To understand the Corbit plot, numbers around the outermost ring correspond to the time lags  $l$ , and the concentric rings denote the stage of neighbors  $r$ , where the innermost ring is for the first-stage neighbors, the

next ring for the second-stage neighbors, etc. Hence, each point's position encodes both its stage of neighbors  $r$  (by which ring it lies on) and its lag  $l$  (by its angular placement on that ring). For example, the point on the innermost ring with angular placement at 1 shows  $\text{nvacf}(1, 1)$ , the next anticlockwise point on the same ring is  $\text{nvacf}(2, 1)$ , and the translational point (still with angular placement 1) on the second innermost ring is then  $\text{nvacf}(1, 2)$ .

Through the Corbit plot, we see that the NVACF shares a pattern similar to that of the NACF, the correlations decreasing both as the time delay  $l$  increases and as one moves to more distant  $r$ -stage neighbors. This aligns with the intuition that volatility shocks lose their influence over time (increase in  $l$ ) and across the network that recent shocks carry stronger autocorrelation,

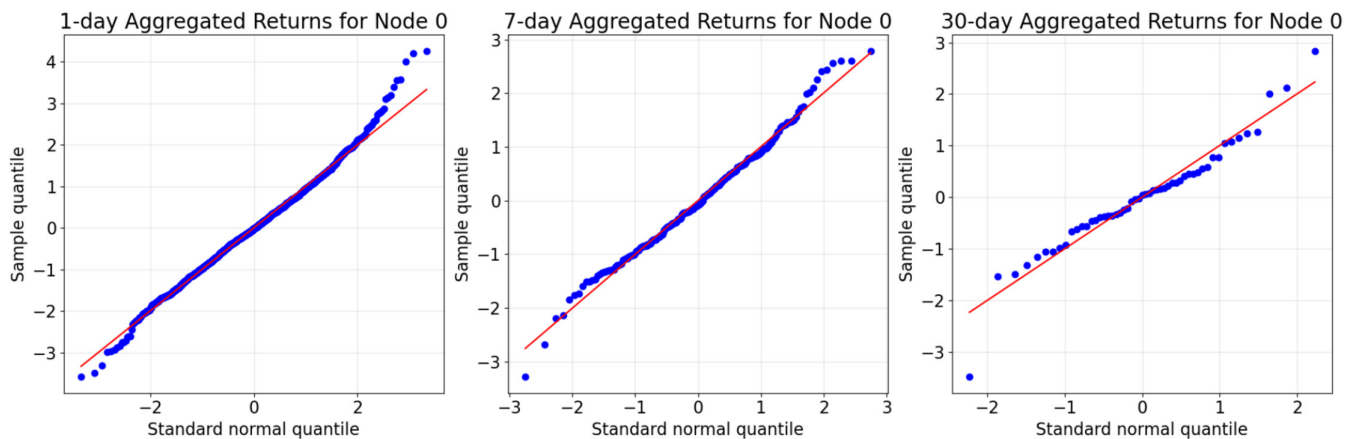


FIGURE 5 | QQ plot for node 0 aggregated returns over days, weeks and months.

DCC-GNGARCH  $\text{nvacf}$  (lags 1 to 20, stages 1 to 3) Corbit plot

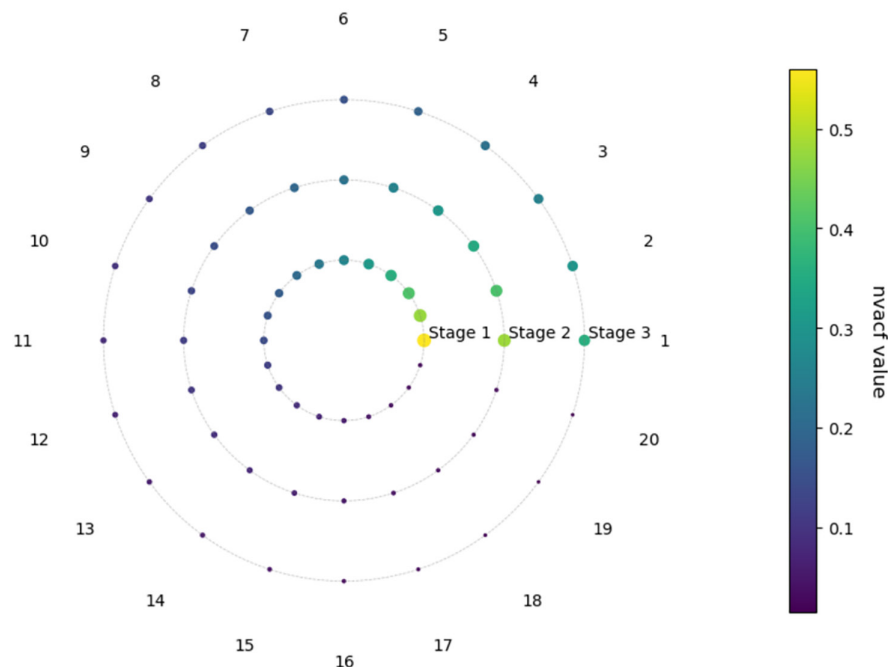


FIGURE 6 | Corbit plot for our simulated data with fitted DCC(1,1)-GNGARCH(1, 1, [1], [1]) model, up to max lag  $l = 20$  and max stage of neighbors  $r = 3$ .

whereas remote or older disturbances (distant  $r$ -stage neighbors) have progressively weaker effects.

### 3.3 | Parameter Estimation

We have so far investigated the model validity via stylized facts and its spatial autocorrelation illustration, indicating a sensible and promising volatility model. An important task now is to “re-fit” the parameters with the known simulated returns, which will allow us to complete the parameter fitting of DCC(1,1)-GNGARCH(1,1,[1],[1]). Our parameter estimation scheme is based on a numerical optimization process with a proxy of latent variable volatility and a certain loss function, then employ a gradient-based optimizer to minimize it. The fitting routine works if the optimizer can recover the ground-truth coefficients robustly on synthetic data. Once validated, we can then apply the same estimation framework to real market return series to complete the parameter calibration of our proposed network-augmented volatility model.

#### 3.3.1 | Volatility/Variance Proxy

While the random variable “volatility” is important for capturing the overall trend and pattern of the financial return series and forecasting, it is inherently latent (Hansen and Lunde 2006) and cannot be observed directly (Tsay 2010b). People therefore invent different volatility models or proxies to estimate volatility as a form of conditional standard deviation, or sometimes equivalently the conditional variance as its squared form.

Inevitably, these volatility or variance proxies are only “imperfect” approximations. In this paper, under our DCC-GNGARCH framework, we use the squared returns at time  $t$  (vectorized as  $\mathbf{X}_t \mathbf{X}_t^T$ ) as a proxy for the conditional covariance matrix, and conduct later estimation and forecast comparison on the variance scale rather than the standard deviation scale. The reason is its conditional unbiasedness with respect to the model conditional variance  $\hat{\Sigma}_t^1$ , whereas even a conditionally unbiased estimate of the conditional variance yields a conditionally biased estimate of the conditional standard deviation once simply taking the square root. We include the proof of the conditional unbiasedness of  $\mathbf{X}_t \mathbf{X}_t^T$  in Proposition B.2, along with the proof of the square root bias in Proposition B.6 by Jensen's inequality, in Supporting Information S2.

Although the squared return provides a simple conditionally unbiased variance proxy, it also suffers from noisy measurements that accumulate between observations, inducing substantial observation-to-observation noise relative to the true conditional variance and consequently worsens the predictive performance of our volatility model (Andersen and Bollerslev 1998). In network structures, this noise intervention becomes even more pronounced once our network is complex with many nodes. Practitioners sometimes introduce better proxies, such as the realized variance with high-frequency intraday data (Patton 2011). Alternatively, when the squared return remains the proxy of choice, different loss functions in numerical optimization can be applied to mitigate the impact of noise on forecast

errors via variance reduction, helping to achieve better forecast performance.

#### 3.3.2 | Loss Function

Another important part of our numerical optimization is the choice of loss function  $L$ , which links the chosen variance proxy and the conditional variance derived from the volatility model. In the DCC-GNGARCH framework, once we set the squared return  $\mathbf{X}_t \mathbf{X}_t^T$  as the variance proxy and  $\hat{\Sigma}_t$  as the model-implied conditional variance, parameter estimation can be formulated by minimizing a loss function  $L(\mathbf{X}_t \mathbf{X}_t^T, \hat{\Sigma}_t)$ .

Although numerous loss functions can be used, Patton (2011) rigorously introduced two important features as a good loss: robustness and homogeneity, and summarized the form of robust and homogeneous loss functions in detail. In this paper, we focus on the two that are most commonly used: MSE and QLIKE function.

**3.3.2.1 | MSE.** Consider we have a sequence of return observations  $\{\mathbf{X}_t\}_{t=0}^{T-1}$  (with  $\mathbf{X}_0$  as the initial return, and overall, we have  $T$  data points), and let  $\{\hat{\Sigma}_t\}_{t=1}^{T-1}$  be the corresponding conditional variance sequence from a certain volatility model, that each  $\hat{\Sigma}_t$  is built off information up to  $\mathcal{F}_{t-1} = \sigma(\mathbf{X}_0, \dots, \mathbf{X}_{t-1})$ , then for a network with total  $d$  nodes, the MSE loss is defined as

$$L_{\text{MSE}}(\mathbf{X}_t \mathbf{X}_t^T, \hat{\Sigma}_t) = \sum_{i=1}^{T-1} \left[ d^{-2} \sum_{i=1}^d \sum_{j=1}^d (\mathbf{X}_t \mathbf{X}_t^T - \hat{\Sigma}_t)_{ij}^2 \right] \quad (12)$$

where the subscript represents the  $ij$ -th entry of the interested matrix. This measure quantifies the discrepancy between the squared return proxy and the predicted variances of the model, and we aim to minimize this loss to capture the accurate volatility forecast from the model.

MSE is regarded as the only loss function that is both robust on the forecast error, set as the difference between variance proxy (using the squared return  $\mathbf{X}_t \mathbf{X}_t^T$  here) and the conditional variance of the model  $\hat{\Sigma}_t$ , and homogeneous (Patton 2011). However, its performance can be less reliable in nonlinear, heteroskedastic settings (Andersen and Bollerslev 1998), since least-squares estimates are sensitive to outliers of the variance proxy (Hansen and Lunde 2006), and when MSE is used to tune a volatility model based on the usual forecast error, the resulting error variance is proportional to the square of the return variance (Patton 2011). These limitations highlight the drawbacks of using the squared return as the conditional variance proxy and MSE as the loss function for model parameter estimation.

**3.3.2.2 | QLIKE/NLL.** Another commonly used parameter estimation scheme is to deduce the QMLE via (7), under the normality assumption. As we usually prefer minimization of the loss function, we naturally apply the NLL as the loss function, which is invariant up to additive constant and an overall scale factor with the well-known QLIKE function.

Because of this, NLL and QLIKE shall share the same property, and according to Patton (2011), QLIKE is the only robust loss function on the standardized forecast error  $\mathbf{X}_t^T \hat{\Sigma}_t^{-1} \mathbf{X}_t$  and is homogeneous. Meanwhile, it is less sensitive to extreme values of  $\mathbf{X}_t \mathbf{X}_t^T$ , with the conditional variance of the standardized forecast error being approximately a constant, while still guaranteeing the “unbiasedness” behavior that the conditional expectation of the standardized forecast error equals 1.

**3.3.2.3 | Shrinkage and Regularization.** When the baseline model is specified as the parsimonious DCC(1,1)-GNGARCH(1,1,[1],[1]) requiring only seven parameters in total, the risk of overfitting due to over-parameterization is rather limited. However, for more general specifications such as DCC( $M,N$ )-GNGARCH( $p,q,[s_1,\dots,s_q],[r_1,\dots,r_p]$ ), the number of unknown parameters may increase substantially, especially when higher order lags, multiple neighborhood stages, or node-specific coefficients are introduced as a local version of the model. In such cases, regularization is suggested to control model complexity and reduce the risk of overfitting, for example, applying shrinkage techniques like Ridge ( $L_2$ ) or Lasso ( $L_1$ ) regularization directly to the loss function.

Letting  $\psi = (\theta, \phi)$  denote the full model parameter vector in the notation of Section 2.2.1, we introduce a regularization penalty  $\xi > 0$  to the original loss function. The regularized loss functions then become

$$L_{\text{ridge}} = L + \xi \|\psi\|_2^2, \quad L_{\text{lasso}} = L + \xi \|\psi\|_1$$

where  $L$  denotes the original loss function (either the MSE defined in (12) or the NLL as the negative version of (7)), and  $L_{\text{ridge}}$  and  $L_{\text{lasso}}$  represent the ridge- and lasso-regularized loss functions, respectively. These penalized loss functions help control model complexity by shrinking redundant parameters to zero, therefore restricting the number of active parameters and mitigating the risk of over-parameterization.

### 3.3.3 | Fitting Results

With the above discussion of variance proxy and loss function, we can now carry the optimization process to fit the specified DCC(1,1)-GNGARCH(1,1,[1],[1]) model parameters, with the adaptive optimizer Adam with learning rate 0.01. Table 2

**TABLE 2** | Parameter estimates (2 d.p.) for simulated returns with seed 0 and 500 epochs.

| Parameter     | True | Estimate (MSE) | Estimate (NLL) |
|---------------|------|----------------|----------------|
| $\alpha_0$    | 0.05 | 0.05           | 0.05           |
| $\alpha_1$    | 0.20 | 0.16           | 0.20           |
| $\gamma_1$    | 0.60 | 0.65           | 0.59           |
| $\beta_{11}$  | 0.05 | 0.01           | 0.04           |
| $\delta_{11}$ | 0.05 | 0.07           | 0.06           |
| $\lambda$     | 0.40 | 0.40           | 0.41           |
| $\omega$      | 0.30 | 0.29           | 0.28           |

reports both the true parameters (as mentioned in (11)) and the estimated coefficients under both loss functions applied to the simulated return dataset with random seed 0 (which is the same data analyzed throughout all previous sections) and 500 epochs, and Figure 7 illustrates the convergence of loss over epochs during the training process.

The drawback of using MSE is reflected in Table 2, where the NLL estimates are uniformly closer to the true parameter values. The training curves shown in Figure 7 further support this observation, that the NLL loss rapidly decreases and stabilizes, while the MSE loss declines more gradually, indicating that the latter is more affected by noise in the realized variance proxy.

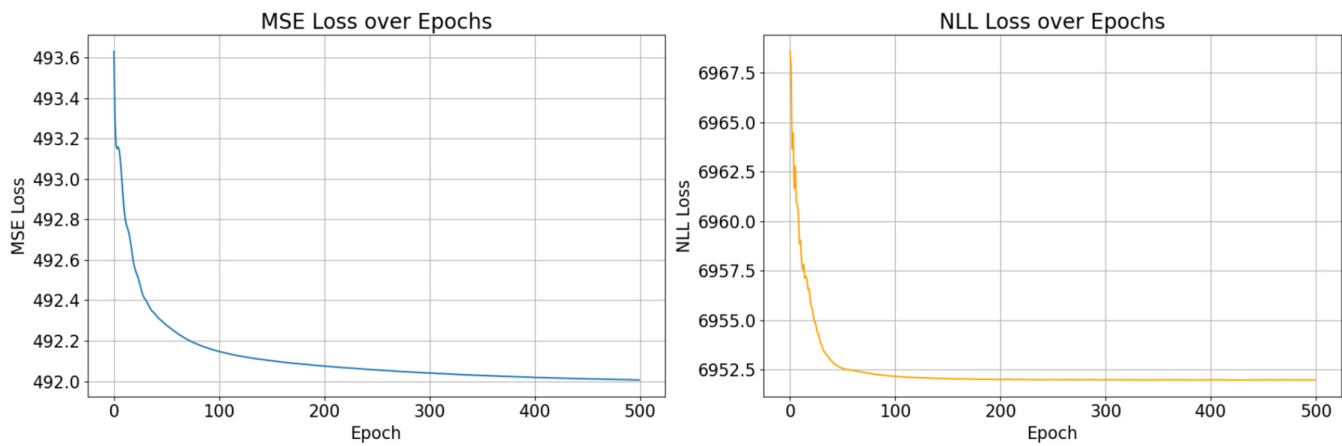
To further evaluate the robustness of parameter estimates, we rerun the optimization for small independently simulated datasets (using different random seeds). Here, we run 20 replications as they achieve a practical balance between computational cost and reliability of our summary statistics. For each run, we record the estimated coefficients and then summarize, see Table 3 below and Boxplot in Supporting Information S3: Figure 13.

We observe that the estimates obtained under the NLL criterion are generally more stable and concentrated around the true parameter values, whereas those obtained under the MSE criterion exhibit greater variability across replications. This is consistent with the previous single-run fitting results, where the NLL estimates were already seen to be closer to the true coefficients. Since the simulated data are generated under the conditional Gaussian assumption, the NLL objective is better aligned with the underlying model structure and therefore yields more statistically efficient parameter estimates. For this reason, in the subsequent empirical parameter estimation, we adopt the QMLE approach as our primary fitting method.

## 4 | Modeling Selected US Stocks

After analyzing the specified DCC(1,1)-GNGARCH(1,1,[1],[1]) on the simulated data, we are now interested to see whether our model can fit the real return series and make reasonable volatility predictions. The return series dataset used for training the model in this section is taken as log-returns from (6b) of Definition 4 derived from the 75 selected US stocks' daily closing price in the market (updated on June 19th, 2025), including well-known stocks like NVIDIA (NVDA), Tesla (TSLA), and Intel (INTC), in a duration from January 1st, 2023, to July 31st, 2025.

Prior to fitting the DCC(1,1)-GNGARCH(1,1,[1],[1]) model to real-world log-returns, we firstly ensure that each log-return series is demeaned prior to estimation, so that the model is fitted to centered returns rather than raw returns. This is in line with the zero-mean specification in (1a–1c) while preserving the main focus on conditional variance and correlation dynamics. Furthermore, we verify that the demeaned log-return series is (first-order) stationary, that ensures we can fit a nonexplosive model with appropriate parameters.



**FIGURE 7** | Training curves for MSE loss (left) and NLL loss (right) against epochs.

**TABLE 3** | Parameter estimates up to 2 d.p., including the position quantities (mean and median), and dispersion quantities (standard deviation and inter-quartile range [IQR]) over seeds from 0 to 19 for the MSE and NLL fitting methods. Overall, both methods recover the true parameter values reasonably well under this simple simulation setting. However, the NLL-based estimates are generally closer to the true parameters, while also showing smaller dispersion for most parameters. These results suggest that, although both criteria perform satisfactorily on simulated data, the NLL approach provides more accurate, less biased, and overall more stable parameter recovery in this experiment.

| Parameter     | True | Mean |      | Median |      | Std. |      | IQR  |      |
|---------------|------|------|------|--------|------|------|------|------|------|
|               |      | MSE  | NLL  | MSE    | NLL  | MSE  | NLL  | MSE  | NLL  |
| $\alpha_0$    | 0.05 | 0.10 | 0.06 | 0.08   | 0.05 | 0.07 | 0.02 | 0.07 | 0.02 |
| $\alpha_1$    | 0.20 | 0.14 | 0.19 | 0.14   | 0.20 | 0.05 | 0.05 | 0.08 | 0.04 |
| $\gamma_1$    | 0.60 | 0.50 | 0.53 | 0.58   | 0.59 | 0.17 | 0.14 | 0.25 | 0.10 |
| $\beta_{11}$  | 0.05 | 0.03 | 0.07 | 0.02   | 0.05 | 0.02 | 0.07 | 0.03 | 0.03 |
| $\delta_{11}$ | 0.05 | 0.08 | 0.07 | 0.04   | 0.05 | 0.13 | 0.10 | 0.06 | 0.06 |
| $\lambda$     | 0.40 | 0.39 | 0.39 | 0.39   | 0.41 | 0.13 | 0.08 | 0.11 | 0.05 |
| $\omega$      | 0.30 | 0.33 | 0.30 | 0.30   | 0.28 | 0.18 | 0.15 | 0.15 | 0.07 |

#### 4.1 | Treatment of the Conditional Mean

As stated, we simply demean the raw return series to ensure zero conditional mean for data in the estimation period. In particular, for each stock  $i$ , we subtract its sample mean from the return series and define the centered return by

$$X_{i,t} = r_{i,t} - \bar{r}_i$$

where  $r_{i,t}$  is the raw, observed return and  $\bar{r}_i$  is the corresponding sample mean. The DCC-GNGARCH model is then fitted to the centered return vector  $\mathbf{X}_i$ . This provides a parsimonious treatment of the conditional mean while allowing the model to focus on the volatility spillover and dependence structure of interest.

#### 4.2 | Stationarity (First-Order)

Stationarity of the log-return series is important for fitting a bounded and well-behaved volatility process, and we assess it by the augmented Dickey–Fuller (ADF) test, originated from the Dickey–Fuller (DF) test (Dickey and Fuller 1979), with the null hypothesis that the process has a unit root and is nonstationary.

Since the log-return (6b) is in the form of a differencing series, strong deterministic trends are not expected (otherwise there is a continuous exponential rise/fall in the price of the stock). For all stocks in the dataset, the ADF tests without trend reject the null hypothesis at the 5% significance level, supporting the use of the proposed DCC(1,1)-GNGARCH(1,1,[1],[1]) model on the demeaned log-returns.

#### 4.3 | Network Construction

Traditional financial networks often rely on correlation-based strategies on price returns and place threshold filtering to define edges between assets, such as the correlation-of-correlation (CoC) method in Tapia Costa et al. (2025)<sup>2</sup>. While a correlation-based threshold network is simple to implement, it is potentially more endogenous because the network is derived from the same return series whose volatility dynamics are subsequently modeled.

To provide a more economically interpretable network specification, we construct the network using stock characteristics rather than relying solely on return co-movement. By defining



nodes through a high-dimensional feature vector, the network maps the fundamental structural characteristics of the assets. Consequently, stocks with similar behavioral profiles naturally cluster together independently of daily market fluctuations, providing a more robust graphical topology for analyzing cross-sectional relationships.

#### 4.3.1 | Resolving Endogeneity and Look-Ahead Bias

An important issue in financial network modeling is that the network specification may itself be influenced by information from the model estimation period, which can give rise to both endogeneity concerns and look-ahead bias.

To strictly prevent future contamination, we apply a static network constructed entirely from a period strictly prior to our training data. The methodology utilizes a nonoverlapping formation window from April 29, 2022 to December 31, 2022, to compute all stocks' characteristics. The resulting network is then kept constant throughout the subsequent out-of-sample training period (beginning January 1, 2023). Specifically, this means for each stock  $i$ , we define  $v_i = (v_i^{(1)}, \dots, v_i^{(p)})^T \in \mathbb{R}^p$  as the characteristic vector, where the components of  $v_i$  are features computed using only information from the formation window, and then use these characteristic vectors to construct the network.

Because no contemporaneous data from the training set is used to define the edges, look-ahead bias is entirely eliminated, and endogeneity concerns regarding the network's structure are substantially reduced by constructing the network from stock characteristics rather than solely on return correlations.

#### 4.3.2 | Feature Selection

To represent each stock by a broad market profile during the formation window, we first apply 11 continuous market-derived characteristics for each stock. These variables are computed entirely from daily price and trading volume data, and therefore avoid the timing problems often associated with quarterly accounting variables. At the same time, they provide a timely summary of how each stock behaves in the market, capturing differences in performance, risk exposure, market sensitivity, and trading activity.

The selected continuous characteristics can be grouped into four main aspects:

- **Return and momentum:** This group summarizes the stock's average performance and directional behavior over the formation window, including (i) mean daily log-return, measuring the stock's average day-to-day appreciation; (ii) mean absolute log-return, reflecting the typical magnitude of daily movement regardless of sign; and (iii) total momentum, defined as the cumulative sum of log-returns, capturing the overall trend of the stock during the formation window and helping distinguish persistently rising stocks from persistently declining ones.

- **Risk and tail behavior:** These variables characterize the variability and downside vulnerability of the stock, rather than serving as conditional variance proxy used later in the DCC-GNGARCH model. Variables including (i) historical return volatility, measured by the sample standard deviation of daily returns over the formation window, which provides a simple summary of the stock's overall realized risk during the formation window; (ii) downside volatility, measured by the sample standard deviation of negative daily returns only, which summarizes the stock's realized exposure to adverse fluctuations during the formation window rather than total variation; (iii) maximum drawdown (MDD), capturing the largest peak-to-trough percentage decline over the formation window, useful for identifying stocks exposed to severe cumulative declines or tail risks.
- **Systematic and idiosyncratic risk:** This group separates market-wide exposure from stock-specific uncertainty, including (i) market beta (estimated relative to the S&P 500 proxy SPY), measuring the sensitivity of the stock to broad market movements; (ii) idiosyncratic volatility (IV), computed from the residuals of the market model, reflecting the portion of return variability that cannot be explained by market exposure alone and therefore capturing stock-specific risk; (iii) correlation between a stock and the market, recording the overall linear association between the stock and the market index, providing an additional summary of co-movement.
- **Liquidity and trading activity:** These features describe how actively and how stably the stock is traded, including (i) average trading volume (in logarithmic form), providing a scale-adjusted measure of typical trading intensity, helping distinguish highly liquid stocks from less actively traded names; and (ii) the volatility of trading logarithmic volume, measuring fluctuations in trading activity over time and can indicate whether the stock experiences stable investor interest or episodic surges in attention and turnover.

To improve the stability of the characteristic matrix, stocks with excessive missing data (exceeding 30% of the continuous feature values) are removed, and any remaining missing entries are imputed by the cross-sectional median of the relevant feature. All continuous features were then standardized feature-wise using Z-score normalization to ensure that no single feature dominates solely because of its numerical scale.

In addition to the continuous variables, we incorporate stocks' sector and industry as categorical features as well. Since such variables are discrete, they are one-hot encoded before entering the feature matrix. The importance of introducing these features is because stocks from the same sector or industry often share common business models, risk exposures and volatility spillover channels, that may not be fully captured by return-based continuous features alone.

Mathematically, if we let  $v_i^{(c)}$  as the standardized continuous feature vector and let  $v_i^{(d)}$  as the one-hot encoded categorical feature vector for stock  $i$ , the combined feature vector is then

$$v_i = \left[ \left( v_i^{(c)} \right)^T, \left( v_i^{(d)} \right)^T \right]^T$$

Given the stock characteristic vectors  $\{v_i\}_{i=1}^n$  over all  $n$  stocks, we compute the pairwise Euclidean distances in terms of

$$d_{ij} = \|v_i - v_j\|_2, \text{ for } i, j = 1, \dots, n$$

Thus, stocks are considered similar when they are close in this characteristic space, vice versa.

#### 4.3.3 | Distance-Threshold Filtering (DTF) Network

We shall obtain a pairwise distance matrix after computing the pairwise Euclidean distance, and our network is then constructed using a DTF rule. Explicitly, given a pre-determined threshold, where the default value is set as the 20% quantile of the upper off-diagonal entries of the pairwise distance matrix, and with  $d_{ij}$  representing the pairwise Euclidean distance between stock  $i$  and  $j$ , the corresponding graph adjacency matrix  $A = (a_{ij})$  is given by

$$a_{ij} = \begin{cases} 1 & \text{if } d_{ij} \leq \text{threshold} \\ 0 & \text{otherwise} \end{cases} \text{ and } a_{ii} = 0$$

Such network yields an economically meaningful notion of neighborhood: Two stocks are linked when they are close in terms of their return behavior, risk profile, liquidity conditions, and business classification, so that direct connections correspond to similarity in their overall market and stock characteristics.

#### 4.3.4 | Network Sparsity and Computational Scalability

In the DTF network construction, the choice of distance quantile affects not only the topology of the stock network but also its sparsity. We quantify the degree of sparsity by using edge density, that is, for an undirected graph with  $d$  nodes, the edge density is defined as

$$\rho = \frac{|E|}{\binom{d}{2}} = \frac{2|E|}{d(d-1)} \quad (13)$$

where  $|E|$  is the number of edges in the network. Edge density serves as a simple measure of the network complexity, which is relevant when assessing possible overfitting and scalability in higher dimensions. Smaller quantile thresholds yield sparser networks and lower edge densities, whereas larger thresholds generate denser graphs, resulting in a greater number of potential spillover interactions and a higher risk of model overfitting.

To empirically validate this sparsity constraint, we evaluate the model's performance across varying network threshold levels: a highly sparse network (e.g., the 0.02 quantile), our baseline moderate network (the 0.20 quantile), and a nearly dense network (the 0.80 quantile). We will carry information criteria for in-sample model preference and related tests on out-of-sample predictive performance later, and the results are concluded in Section 4.9.

### 4.4 | Economic Interpretation of Model Spillover Parameters

The spillover coefficients in the DCC-GNGARCH model are the network effect parameters  $\beta_{kr}$  and  $\delta_{\ell r'}$  in (1b), for  $k = 1, \dots, q$ ,  $\ell = 1, \dots, p$ ,  $r = s_1, \dots, s_q$  and  $r' = r_1, \dots, r_p$ . For the DCC(1,1)-GNGARCH(1,1,[1],[1]) specification, these are reduced to  $\beta_{11}$  and  $\delta_{11}$ . In the financial network setting, these spillover parameters have a clear economic interpretation.

The coefficient  $\beta_{kr}$  measures how strongly recent shocks in neighboring stocks in the network, in terms of lagged squared returns, contribute to the current conditional variance of a given stock. A larger value of  $\beta_{kr}$  indicates stronger volatility transmission across connected firms. Meanwhile, the coefficient  $\delta_{\ell r'}$  measures how strongly the past conditional variances of neighboring stocks in the network persist into the current conditional variance of a given stock and can therefore be interpreted as a network persistence effect. Since our model parameters are set under the non-negativity constraints (8), these network spillover effects are interpreted as amplifying channels of volatility propagation rather than offsetting ones.

In the present characteristic-based network, first-order neighbor effects are economically interpretable as direct linkages between stocks with similar return dynamics, risk characteristics, market exposure, liquidity conditions, and common industry or sector classification. More broadly, such links can reflect exposure to similar economic drivers, investor clientele, or financial information environments. If higher order neighbor terms are included, they capture more indirect forms of propagation, where volatility spreads through intermediate nodes and reflects broader sectoral or market-wide transmission rather than immediate pairwise similarity.

Although our DCC-GNGARCH model uses global spillover coefficients as discussed in (5) of Section 2.2, the importance of network transmission still differs across assets, because highly connected stocks can both receive and transmit volatility through more channels than less connected stocks. In this sense, even under common global parameters, the model permits heterogeneous effective spillover exposure across prominent firms and subnetworks.

### 4.5 | Model Fit

With the constructed network, we now apply the fitting scheme in Section 3.3, with squared return as the variance proxy and QLIKE/NLL loss, to fit a DCC(1,1)-GNGARCH(1,1,[1],[1]) model on the dataset. Table 4 records the parameter estimates.

Because our dataset uses daily closing prices, the resulted returns include overnight jumps and tend to be noisier. To reduce this effect, Hansen and Lunde (2005) introduced a rescaling technique that matches the level of volatility forecasts to squared returns. Their approach of rescaling involves the realized variance, which is defined as the sum of squared intraday returns for high-frequency data. However, since our data are daily returns, the realized variance is then approximated as the squared return on this specific day. We adapt this idea but implement a bit

differently for dynamic rescaling, by aligning the level of model conditional variances with observed, centered squared returns, dynamically within a rescale window (with default window size  $n_w = 252$  for approximate number of trading days in a year). To formulate, for stock  $i$ , its rescaled conditional variance is

$$\tilde{\sigma}_{i,t} = c_{i,t} \hat{\sigma}_{i,t}, \quad c_{i,t} = \begin{cases} \frac{n_w^{-1} \sum_{s=t-n_w+1}^t r_{i,s}^2}{n_w^{-1} \sum_{s=t-n_w+1}^t \hat{\sigma}_{i,s}^2} & \text{when } t \geq n_w \\ \frac{n_w^{-1} \sum_{s=1}^{n_w} r_{i,s}^2}{n_w^{-1} \sum_{s=1}^{n_w} \hat{\sigma}_{i,s}^2} & \text{when } t < n_w \end{cases} \quad (14)$$

where  $\hat{\sigma}_{i,t} = (\hat{\Sigma}_t)_{ii}$  as the model-implied conditional variance for stock  $i$  at time  $t$ ,  $r_{i,s}$  is the daily log-return for stock  $i$  at time  $s$ , and  $n_w = 252$  for the yearly rescale window. According to Patton (2011), since the QLIKE loss functions are homogeneous, we still have ranking consistency after rescaling.

**TABLE 4** | Parameter estimates up to 4 d.p., training with 1000 epochs.

| Parameter     | Estimate (NLL) |
|---------------|----------------|
| $\alpha_0$    | 0.0005         |
| $\alpha_1$    | 0.5995         |
| $\gamma_1$    | 0.3195         |
| $\beta_{11}$  | 0.0758         |
| $\delta_{11}$ | 0.0051         |
| $\lambda$     | 0.1467         |
| $\omega$      | 0.0349         |

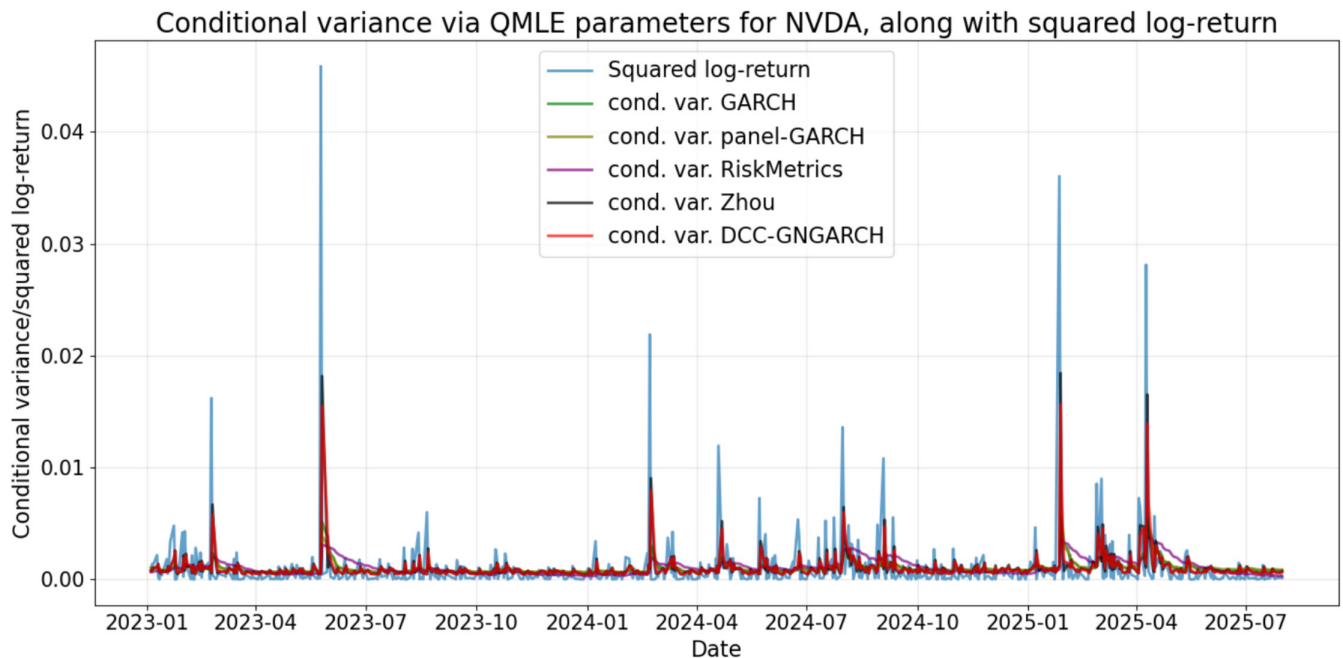
Figure 8 provides the squared log-returns and conditional variances under four different models: (i) the best fitting univariate GARCH(1,1) model; (ii) a panel-GARCH(1,1) specification following Pakel et al. (2011) and Su and Zhu (2024), assuming zero-mean Gaussian innovations. This model controls for individual asset heterogeneity by assigning a stock-specific fixed effect,  $\omega_i$ , to the variance equation, while pooling the dynamic persistence parameters ( $\alpha$  and  $\beta$ ) across all stocks. For stock  $i$ , this panel-GARCH(1,1) is written as

$$X_{i,t} \sim \mathcal{N}(0, \sigma_{i,t}^2) \\ \sigma_{i,t}^2 = \omega_i + \alpha X_{i,t-1}^2 + \beta \sigma_{i,t-1}^2$$

with  $\omega_i > 0$ ,  $\alpha, \beta \in [0, 1)$  and  $\alpha + \beta < 1$  to ensure stationarity and ergodicity. Although  $\omega_i$  is an unobservable fixed effect (Su and Zhu 2024), according to Pakel et al. (2011), it can be estimated by variance tracking/targeting estimation (VTE) that

$$\omega_i = \gamma_i(1 - \alpha - \beta), \quad \gamma_i = E(X_{i,t}^2)$$

Based on the properties of stationarity and ergodicity (Hamilton 1994), we can apply the method of moments (MM) to estimate the unconditional variance as  $\hat{\gamma}_i = \frac{1}{T} \sum_{t=1}^T X_{i,t}^2$ , and we will carry QMLE optimization to estimate  $\alpha$  and  $\beta$  for the panel-GARCH(1,1), which are given as  $\alpha = 0.0877, \beta = 0.7528$ ; (iii) the RiskMetrics procedure (Patton 2011) defined as  $\hat{\sigma}_{i,t}^2 = \lambda \hat{\sigma}_{i,t-1}^2 + (1 - \lambda) r_{i,t-1}^2$ , with  $\lambda = 0.94$ ; (iv) Zhou's network GARCH model (Zhou et al. 2020), estimated by QMLE (using the fitted parameters  $\omega = 0.0007, \alpha_0 = 0.6785, \lambda_0 = 0.1121, \beta_0 = 0.1073$  under the notation of that paper); finally (v) our QMLE-fitted DCC(1,1)-GNGARCH(1, 1, [1], [1]), for the stock NVDA during the training period.



**FIGURE 8** | Conditional variance comparison for NVDA across models, between the best fit univariate GARCH (green), panel-GARCH(1,1) (olive), RiskMetrics ( $\lambda = 0.94$ , purple), QMLE-fitted Zhou's network GARCH model (black), and DCC(1,1)-GNGARCH(1, 1, [1], [1]) (red).

Since squared returns are our variance proxy, we expect they share a similar pattern with model conditional variances. From Figure 8, we first observe intermittent and sharp spikes for the example stock NVDA's squared log-returns, particularly around February 2023, May 2023, February 2024, May 2024, August 2024, September 2024, February 2025, and May 2025. Compared with best fit univariate GARCH(1,1), panel-GARCH(1,1), and RiskMetrics, our fitted DCC-GNGARCH tracks those spikes more accurately, producing visible and intense local peaks that align with the large, realized spikes; when we further compare the fitted DCC(1,1)-GNGARCH(1,1,[1],[1]) model with the Zhou's network GARCH model, both models capture the major volatility bursts in NVDA and produce pronounced local peaks around the largest realized movements. Overall, the two fitted conditional variance paths are quite similar over most of the sample, which is desirable as the proposed DCC(1,1)-GNGARCH(1,1,[1],[1]) generally following Zhou's model closely in conditional variance, with newly added persistence effect.

On the other hand, the model conditional covariance is often used to estimate the latent (squared) co-volatility (Asai and McAleer 2015), and the cross-product can be analogously seen as a noisy proxy. Since our DCC-GNGARCH includes time-varying covariances, we can investigate pairwise model conditional covariances (after the dynamic rescaling (14) with rescale window  $n_w = 252$  by replacing squared returns and conditional variance with cross products and conditional covariance, respectively) for two stocks and compare them to the cross-product of corresponding log-returns. To act as a standard with respect to the zero baseline in the centered cross product of stock prices, we implement the rolling median of estimated model conditional covariance as the standard measure, such that the model conditional covariance higher than this rolling median is set to have a positive co-movement, vice versa. We include both in Figure 9, and comparing the two therefore provides a direct diagnostic

of how well the model captures dynamic co-movements. In fact, our fitted DCC(1,1)-GNGARCH(1,1,[1],[1]) is again able to track the spikes on the training set very well, but also some troughs (for example, on May-June 2024 approximately), although the trough tracking is not so obvious and easy to detect.

#### 4.6 | In-Sample Model Comparison

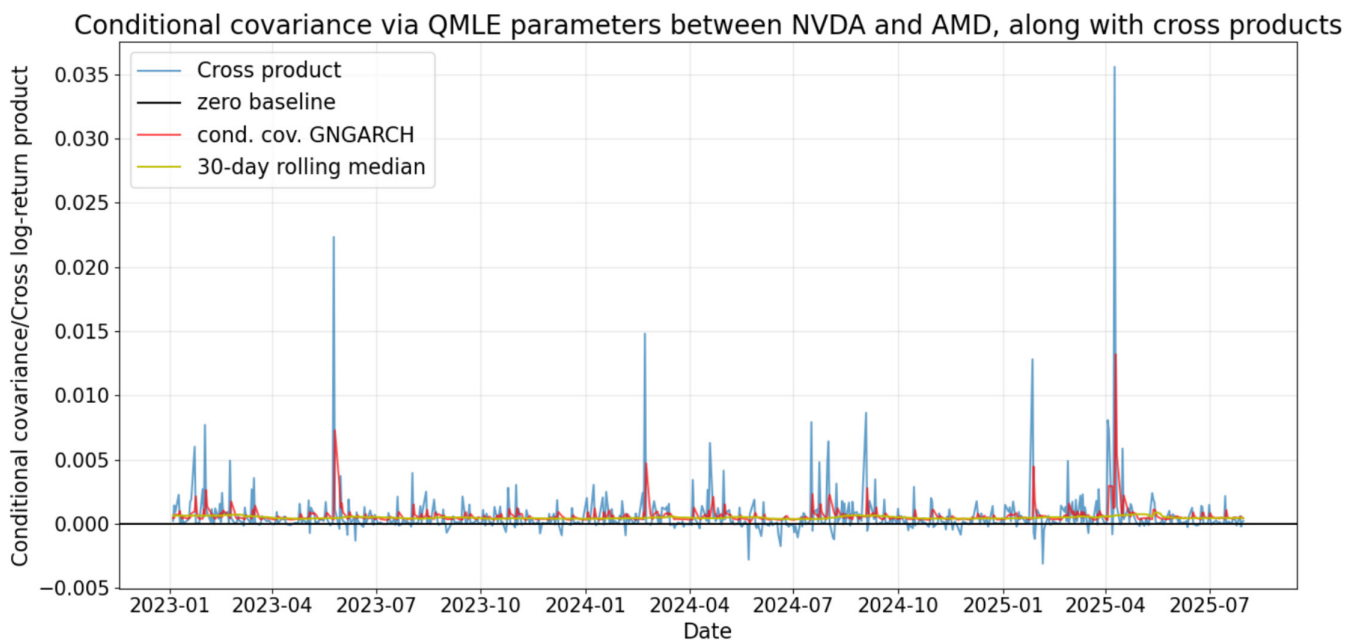
To formally evaluate which specification best captures the underlying volatility and correlation dynamics of the asset network, we conduct an in-sample model comparison using the full training dataset. As emphasized in Diebold (2015), pseudo-out-of-sample testing is often "abused" to determine which model is the "truest" representation of the data-generating process, and he argued that for model comparison, researchers should apply full-sample procedures, such as information criteria, whereas related Diebold-Mariano (DM) test should be carried only for forecast comparison and predictive accuracy, which will be discussed in next section.

Following this framework, we evaluate our models using the Akaike Information Criterion (AIC, as defined in (15)) and the Bayesian Information Criterion (BIC, as defined in (16)), where  $k$  represents the number of parameters for the model,  $\log L$  is the maximum log-likelihood of the model, and  $T$  is the number of observations. Both criteria resolve the fundamental trade-off between goodness-of-fit and model parsimony by rewarding maximized log-likelihood while penalizing the addition of parameters.

$$AIC = 2k - 2 \log L \quad (15)$$

$$BIC = k \log T - 2 \log L \quad (16)$$

For large sample sizes, the AIC commonly suffers in-sample overfitting (Burnham et al. 2002), as its penalty for model complexity ( $2k$ ) is fixed regardless of the sample size; hence, the AIC



**FIGURE 9** | Cross-product (blue) of log-returns over time between stock NVDA and AMD with zero baseline (black), together with conditional covariance (red) between them, plotted against its 30-day rolling median (yellow).



will usually favor over-parameterized models. By contrast, the BIC generally alleviates overfitting effect much more, by taking the sample size  $T$  into account (with the penalty term  $k \log T$ ), so when a larger sample size naturally results in a larger maximum log-likelihood, we also expect to see a rise in  $k \log T$ , providing a dynamic safeguard against unwarranted model complexity.

Also note that for the panel-GARCH(1,1), multidimensional DCC-GNGARCH and Zhou's network models, the maximum log-likelihood is retrieved directly from the QMLE optimization process, calculated as the negative of the minimized NLL loss after 1000 epochs. In contrast, the baseline univariate GARCH(1,1) specification is estimated individually for each of the 75 stocks. By assuming no volatility spillovers or cross-sectional dependencies, these single-asset models are treated as mutually independent. Consequently, the joint maximum log-likelihood for the entire 75-asset system is computed by simply summing the log-likelihoods of the individually fitted models. Following this assumption of independence, the aggregate AIC and BIC are similarly obtained by summing the respective individual criteria, providing a unified baseline reference for comparison against the multivariate frameworks.

Table 5 presents the evaluation metrics for the full-sample. For both information criteria, a lower (more negative) value indicates superior model preference. The proposed DCC-GNGARCH framework decisively outperforms both the baseline univariate GARCH system, VTE panel-GARCH and Zhou's network model, by attaining the lowest score both under the AIC and BIC. Such result then reflects that the cross-sectional dependencies, neighboring volatility clustering and persistence, and dynamic correlations modeled by our proposed DCC-GNGARCH framework are genuine structural features of the data, rather than the by-product of overfitting.

4.7 | Out-of-Sample Model Forecast Accuracy and Regression

We finally use the constructed DTF network (Figure 10) together with the QMLE-fitted DCC-GNGARCH model to generate volatility and co-volatility forecasts over the out-of-sample period from August 1, 2025 to February 28, 2026, which serves as our test dataset. The proposed model is compared with benchmarks as stated in Section 4.5.

TABLE 5 | Model comparison via AIC/BIC for all 75 stocks.

| Model                            | No. of parameters   | AIC          | BIC          |
|----------------------------------|---------------------|--------------|--------------|
| Best fit GARCH(1,1)              | $3 \times 75 = 225$ | − 200999. 58 | − 199994. 00 |
| VTE Panel-GARCH(1,1)             | 2                   | − 199387. 42 | − 199378. 48 |
| DCC(1,1)-GNGARCH(1, 1, [1], [1]) | 7                   | − 210265. 62 | − 210234. 34 |
| Zhou's network model             | 4                   | − 180047. 25 | − 180029. 37 |

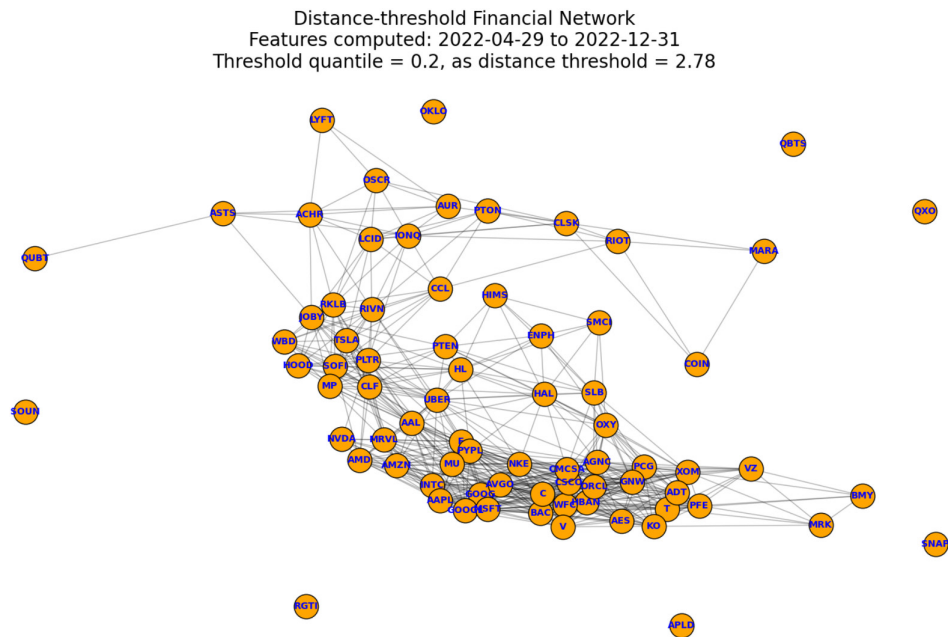


FIGURE 10 | Constructed stock DTF network (edge weights are the pairwise Euclidean distances between connected stocks), with threshold quantile = 0.2 of all the computed stock feature vector pairwise distances in the upper off-diagonal pairwise distance matrix as the default choice. Node names represent the stock name abbreviations.

To evaluate this, we take the one-step ahead forecasting scheme across all models. This means that if we denote the last day of the training period as  $T$ , then we will apply fitted models to predict the conditional variance on the next trading day  $T + 1$  (which is actually the first date in the test period), and then update the observed true log-return at  $T + 1$  from the test set to further perform the one-step ahead conditional variance forecast at  $T + 2$ . The scaling technique mentioned in Section 4.5 is again needed to reduce the noise effect, but the rolling window slides from the training set to the test set, i.e., we will rescale the first forecast with the last  $n_w$  data in the training set, and rescale the second forecast with the last  $n_w - 1$  data in the training set together with the first test set log-return observation, etc. To provide an initial qualitative assessment, we visually inspect the one-step ahead forecasting trajectories generated by the evaluated models across this test dataset, see Figures 11 and 12.

In addition, we will focus on two quantitative aspects for the forecasts from different models. Firstly, we evaluate predictive accuracy through loss function-based comparisons and DM test (Diebold and Mariano 2002), then we evaluate directional movements and trend tracking via the Mincer-Zarnowitz (MZ) test (Mincer and Zarnowitz 1969), with particular attention to the goodness-of-fit measured by  $R^2$ . These two perspectives are useful because a model may perform well in terms of average forecast loss while still differing in how closely it tracks the direction and magnitude of realized volatility movements.

In previous sections, we have used the squared returns as the proxy for the conditional variance. However, since squared returns are highly noisy and contain many isolated spikes, direct qualitative, visual comparison at the daily level can be

misleading. To reduce this noise effect, we here consider a 30-day rolling variance proxy, that is, for stock  $i$ , we define the 30-day rolling realized variance proxy by

$$\widehat{RV}_{i,t}^{(30)} = \frac{1}{30} \sum_{j=0}^{29} X_{i,t-j}^2 \quad (17)$$

whenever the required observations are available. Correspondingly, we shall also take the 30-day rolling conditional variance from our one-step ahead forecasts.

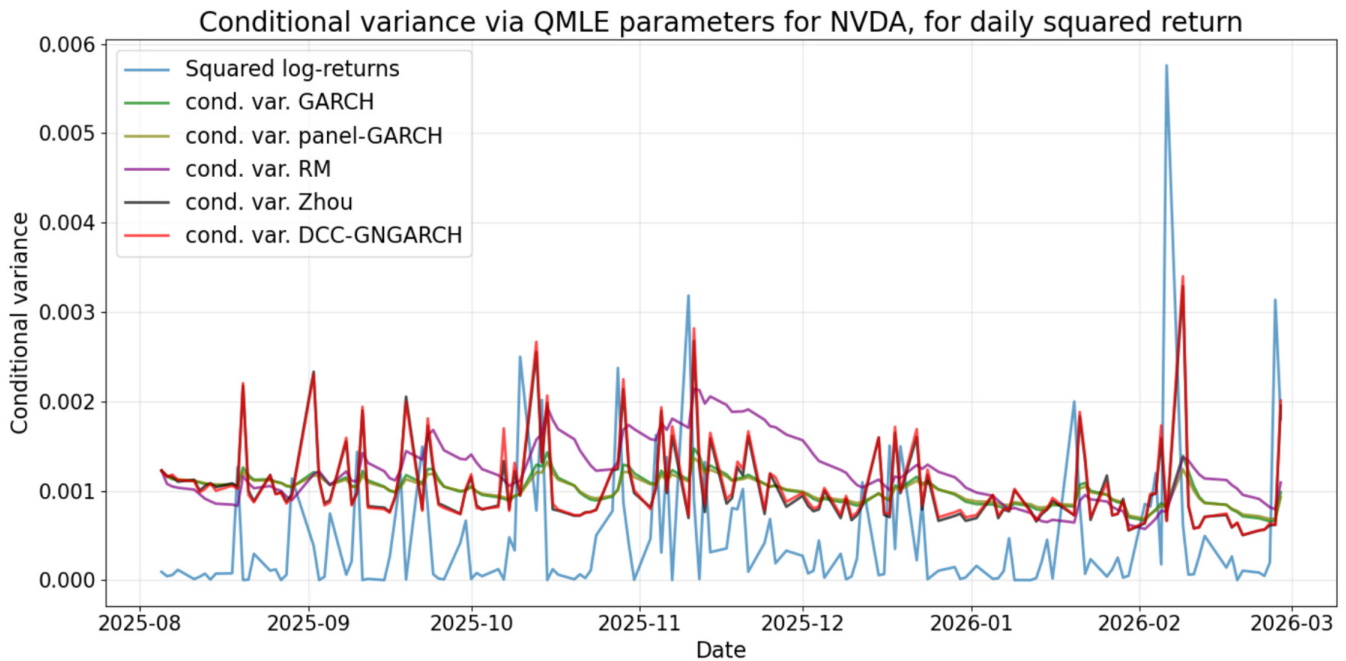
#### 4.7.1 | Forecast Accuracy via DM Test

As mentioned in Diebold (2015), the DM test is, and always should be, a test for “comparing forecasts, not models.” The DM test evaluates the null hypothesis of equal predictive accuracy between two competing forecast series over a specific out-of-sample period, and here, we shall apply the one-sided DM test for claiming the alternative hypothesis be “our fitted DCC-GNGARCH model gives better prediction.”

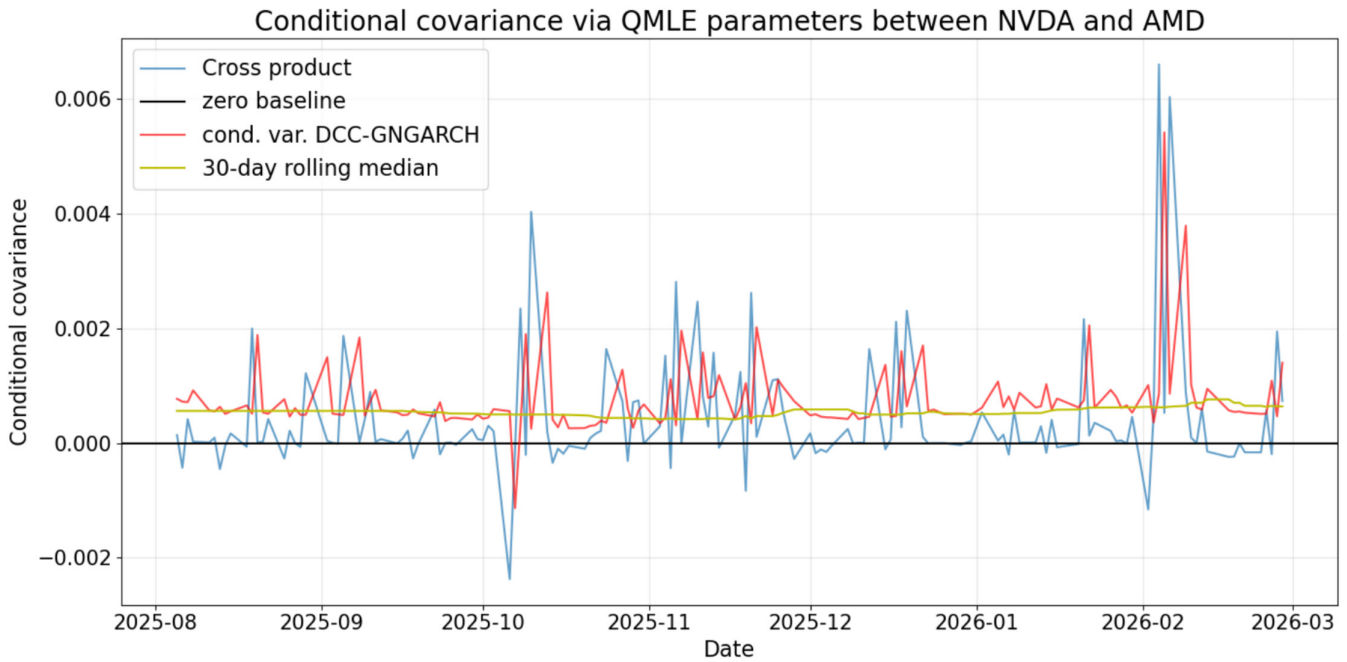
Under our context, let  $y_t = \widehat{RV}_{i,t}^{(30)}$  be the true realization and let  $\hat{y}_{1,t}$  and  $\hat{y}_{2,t}$  denote two model forecasts on conditional variance. Given a specific loss function  $L$ , if  $e_{1,t} = y_t - \hat{y}_{1,t}$  and  $e_{2,t} = y_t - \hat{y}_{2,t}$  being as two models' forecast errors, the loss differential at time  $t$  is defined as

$$d_t = L(e_{1,t}) - L(e_{2,t})$$

and we denote  $\bar{d} = \frac{1}{T'} \sum_{t=1}^{T'} d_t$  is the sample mean of the loss differentials in the test period of length  $T'$ . The null hypothesis of equal



**FIGURE 11** | Out-of-sample conditional variance forecasts for NVDA over the test period (August 2025 to February 2026), comparing the daily squared log-returns against the volatility predictions generated by the proposed DCC-GNGARCH model (red) and the selected benchmark models: standard GARCH (green), panel-GARCH (olive), RiskMetrics (purple), and Zhou's network model (black). Similar with Figure 8, we still generally see that network-based models can track the peaks more sharply and sensitively.



**FIGURE 12** | Out-of-sample conditional covariance forecasts between NVDA and AMD over the test period, with the cross-product (blue) of log-returns over time between stock NVDA and AMD, the zero baseline (black), together with conditional covariance (red) between them, plotted against its 30-day rolling median (yellow). It indicates that the QMLE-fitted model captures the main peaks and troughs reasonably well on the test set, adjusting over time to changes in cross-sectional market co-movements.

predictive accuracy states that the expected loss differential is zero, i.e.,  $H_0: E(d_t) = 0$ . The DM test statistic is constructed as

$$DM = \frac{\bar{d}}{\sqrt{\hat{V}(\bar{d})}}$$

where  $\hat{V}(\bar{d})$  is a consistent estimator of the asymptotic variance of  $\bar{d}$ .

Because financial forecast errors frequently exhibit serial correlation, we evaluate  $\hat{V}(\bar{d})$  using heteroskedasticity and autocorrelation consistent (HAC) estimator (Newey and West 1987), meaning that

$$\begin{aligned} \hat{V}(\bar{d}) &= \frac{1}{T'} \left( \hat{\gamma}_0 + 2 \sum_{k=1}^m w_k \hat{\gamma}_k \right), \quad \hat{\gamma}_k \\ &= \frac{1}{T'} \sum_{t=k+1}^{T'} (d_t - \bar{d})(d_{t-k} - \bar{d}), \quad w_k = 1 - \frac{k}{m+1} \end{aligned}$$

where  $m$  is usually set to the rounded integer of  $T'^{1/3}$  in practice (Zaiontz 2026),  $\hat{\gamma}_k$  is the sample autocovariance at lag  $k$ , and  $w_k$  represents the Bartlett weights.

The DM-HAC statistics for the one-sided test with  $\hat{y}_{1,t}$  being as the rescaled one-step ahead forecast from fitted DCC(1,1)-GNGARCH(1,1,[1],[1]) is concluded in Supporting Information S3: Table 6, with two robust and homogeneous losses QLIKE and MSE (Patton 2011). A negative value of DM-HAC statistic implies a favor of the DCC(1,1)-GNGARCH(1,1,[1],[1]) model, and the DM-HAC statistic less than  $-1.645$  indicates a rejection of the null hypothesis at the 5% significance level, marked with an asterisk. As stated by Patton and Sheppard (2009) and

Patton (2011), the QLIKE loss function yields higher statistical power in DM tests compared to MSE, providing its superiority in volatility forecasting as discussed in Sections 3.3.2 and 3.3.3, although the resulting DM-HAC statistics can still be sensitive to the chosen loss function when evaluating imperfect forecasts.

Supporting Information S3: Table 6 reveals that, in terms of predictive accuracy, our DCC(1,1)-GNGARCH(1,1,[1],[1]) model overall provides a smaller average loss compared with the baseline models' forecasts. Specifically, when evaluated using the preferred QLIKE loss, our proposed model outperforms (with negative value) the best fit univariate GARCH(1,1) and RiskMetrics baselines in 63% (47 out of 75) and 56% (42 out of 75) of the assets, respectively, where 41% (31 out of 75) and 23% (17 out of 75) are statistically significant at the 5% level.

We then evaluate our model against the VTE panel-GARCH(1,1) specification. While the panel-GARCH(1,1) proves to be a highly competitive benchmark, our network-based model still achieves a lower forecast loss with negative DM statistic in roughly half of the panel: 45% (34 out of 75) under QLIKE and 47% (35 out of 75) under MSE. More importantly, when the DCC-GNGARCH model outperforms the panel-GARCH, it does so with striking statistical significance. Among the 34 (35) assets that show a negative DM statistic under the QLIKE (MSE) loss function, 27 (26) significantly reject the null hypothesis of equal predictive accuracy. These statistically significant improvements represent 36% (35%) of the entire 75-stock panel, respectively. This indicates that, although fixed effects are powerful for baseline heterogeneity, explicitly modeling topological risk spillovers provides decisive forecasting advantages for a substantial subset of assets in the network.

Furthermore, the advantage of incorporating neighboring persistence and dynamic conditional correlations is visible when compared to Zhou's network GARCH model. Under both QLIKE and MSE loss functions, the QMLE-fitted DCC(1,1)-GNGARCH(1,1,[1],[1]) model achieves lower average forecasting loss for roughly 70% of the panel, generating statistically significant results for 45% and 52% of the assets, respectively, to QLIKE and MSE loss.

To conclude, these results confirm that the spatial dependencies captured by our network topology, together with the neighboring persistence and dynamic conditional correlation, successfully enhance out-of-sample predictive accuracy across the majority of the chosen stocks.

#### 4.7.2 | Regression Fit via MZ Test

Besides the relative predictive accuracy evaluated by the DM test, the MZ test evaluates the forecast rationality and the directional tracking ability of each individual model and also allows us to assess how well the structural shape of the model forecasts correlate with actual market dynamics over the out-of-sample period.

The MZ regression test is specified as

$$y_t = \alpha + \beta \hat{y}_t + e_t$$

as mentioned in the DM part, we have let  $y_t = \widehat{RV}_{i,t}^{(30)}$  be the actual realization of the unobserved conditional variance, and  $\hat{y}_t$  denote the corresponding 30-day rolling conditional variance forecast from a specific model, with  $e_t$  being as the forecast error. The null hypothesis of perfect forecast calibration states that the forecast is unbiased and perfectly proportional, i.e.,  $H_0: \alpha = 0, \beta = 1$ . However, since squared daily return is a very noisy proxy for true latent squared volatility, regressions between the rolling aggregate and the rolling conditional variance forecast also suffer from these noisy observations to result very small  $p$  values rejecting the null hypothesis and give imperfect forecasts (Patton 2011). While we would also include the values of  $\alpha$  and  $\beta$ , our main focus is on the coefficient of determination ( $R^2$ ), which measures the goodness-of-fit that how well the forecast captures the overall shape of the rolling realized squared volatility series.

The MZ test  $\alpha, \beta$  and  $R^2$  values for the rescaled out-of-sample forecasts are presented in Supporting Information S3: Table 7. A higher  $R^2$  value implies that the model captures the timing of the rise and the fall of volatility more accurately. For clarity, the highest  $R^2$  value for each stock is marked in bold with an asterisk, and we also count the number of largest  $R^2$  value for each model over all 75 stocks. While the best fit GARCH(1,1), VTE panel-GARCH(1,1), RiskMetrics, and Zhou's network GARCH models only provide superior fit in 19%, 8%, 12%, and 20% of the stocks, respectively, our QMLE-fitted DCC(1,1)-GNGARCH(1,1,[1],[1]) model achieves the highest  $R^2$  for 31 out of 75 stocks (41%), which is significantly superior than the other four benchmarks. These results from the MZ test indicate that incorporating the spatial network topology, along with neighboring persistence and dynamic conditional correlations, allows the forecasts to overall track the true trajectory of market volatility more robustly across the majority of the panel.

## 4.8 | Robustness Checks

We finally assess whether our previous conclusions depend strongly on the chosen network specification. To do this, we perform robustness checks across alternative network constructions and threshold choices, along with different choices of rolling window for the realized variance (17) reference, including DTF, CoC, and  $k$ -nearest-neighbor ( $k$ -NN) networks. Related results are shown in Supporting Information S3: Tables 8–10. Briefly speaking, we construct our CoC network (Tapia Costa et al. 2025) by first computing monthly sample correlation matrices of stock log-returns, then taking absolute correlations to focus on dependence strength rather than direction, averaging these matrices across months to form an integrated correlation matrix, and finally filtering this matrix with a set threshold/quantile to determine the graph adjacency structure. On the other hand, the  $k$ -NN network is still constructed from the characteristic-based pairwise distance matrix as used in the DTF, but by linking each stock to its  $k$  closest neighbors.

Supporting Information S3: Table 8 shows that the QMLE-fitted DCC(1,1)-GNGARCH(1,1,[1],[1]) model consistently achieves lower AIC and BIC values than the best fit GARCH(1,1), VTE panel-GARCH(1,1) and Zhou's network model across all networks considered. This indicates that the in-sample model preference of the proposed DCC-GNGARCH is not tied to a single network construction.

Supporting Information S3: Table 9 reports robustness results based on the DM test under both QLIKE and MSE losses, using 30- and 60-day rolling realized variance as the proxy target. Although these forecast results are more mixed than those based on information criteria, they still support the overall competitiveness of the proposed DCC-GNGARCH specification in a range of network constructions and proxy choices. Particularly, the DCC-GNGARCH model tends to outperform Zhou's network GARCH model for a great proportion of stocks, especially under the  $k$ -NN constructions, where the number of stocks with negative DM-HAC statistics is frequently above 50 and in some cases exceeds 60. Relative to the best fit GARCH(1,1) and RiskMetrics benchmarks, the DCC-GNGARCH model also shows favorable performance under several DTF and CoC specifications, particularly under MSE loss, where a majority of stocks exhibit negative DM-HAC statistics, although generally gains are more moderate.

On the other hand, comparing with the VTE panel-GARCH(1,1) benchmark, it is not immediately obvious that our DCC-GNGARCH model achieves superior performance across the full panel of stocks, as the total number of negative DM-HAC statistics is visibly lower. This is particularly evident under the  $k$ -NN networks, where the proposed DCC-GNGARCH model only results negative DM-HAC statistics against panel-GARCH(1,1) for roughly 20–31 stocks, across both rolling windows and loss functions. However, a closer examination of the DTF and CoC networks reveals a more encouraging dynamic. Under these topologies, the DCC-GNGARCH model remains highly competitive, successfully yielding a lower forecasting loss for approximately half the panel (typically ranging between 32 and 40 stocks). More importantly, when the DCC-GNGARCH model results



a lower average loss than the panel-GARCH(1,1), these improvements are overwhelmingly statistically significant as we have discussed in Section 4.7. This pattern is consistent with our previous finding, and again verifies that, although VTE panel-GARCH(1,1) already serves as a parsimonious baseline for general market volatility, for a certain subset of assets, the explicit inclusion of spatial dependencies and dynamic conditional correlations captures risk spillovers that the panel-GARCH misses, and delivers forecasting advantages.

Finally, this robustness check via the DM test also highlights the framework's sensitivity to the chosen network topology. While the  $k$ -NN construction may not be the most suitable structure when compared against baselines like the panel-GARCH(1,1), it performs highly effective when evaluated against contemporary spatial benchmarks such as Zhou's network GARCH model. Meanwhile, the DTF constructions appear to offer a more balanced choice, yielding both economically meaningful network linkages and relatively strong predictive outperformance for the DCC-GNGARCH model.

Supporting Information S3: Table 10 records the counts and proportions of stocks for which each competing model attains the largest MZ  $R^2$  value with different rolling window sizes for the realized variance proxy and under alternative network constructions and threshold choices. From it, we observe that the QMLE-fitted DCC(1,1)-GNGARCH(1,1,[1],[1]) model attains the highest MZ  $R^2$  across stocks for both 30- and 60-day rolling realized variance under every network specification, where the strongest MZ results occur under the DTF network, whereas the CoC and  $k$ -NN versions still remain competitive. This again solidifies our statement in the previous section that the proposed DCC-GNGARCH model is better in describing the pattern of volatility fluctuations, and such advantage is robust to the alternative network specifications.

Throughout the previous analysis, we have always assumed the conditional normality for the innovation across different models, and the fitting is based on QMLE framework. Although, as mentioned in Section 2.2.1 and Bollerslev and Wooldridge (1992), the Gaussian QMLE can achieve consistent and asymptotically normal estimates of model parameters even when the true innovations are non-Gaussian, provided the conditional mean and variance are correctly specified and hold standard regularity conditions. Intuitively, another aspect of robustness check can be performed on the distribution of innovations, where we may set the error in heavy-tailed distribution, like Student- $t$ , and evaluate whether such heavy-tailed error assumption alters our previous findings. Specifically, we investigate whether the relative predictive superiority of the proposed DCC-GNGARCH model holds when all competing models are dynamically equipped to handle heavy-tail innovations. To assess this, we set a constant degree of freedom (default as  $\nu = 6$ ) for the Student- $t$  distribution, ensuring that variations in the error tail-thickness do not act as an additional parameter that affect the comparison between models. For convenience, here, we only evaluate and compare based on the specific DTF network constructed at the 0.20 quantile threshold, and compare the out-of-sample forecasting performance of the competing models under this heavy-tailed assumption. The conclusions are summarized in Supporting Information S3: Tables 11–13.

Specifically, we still observe more negative AIC and BIC values for the fitted DCC(1,1)-GNGARCH(1,1,[1],[1]) model under both Gaussian and Student- $t$  distributed errors in Supporting Information S3: Table 11, again showing robustness in the in-sample model preference. This suggests that the superiority of in-sample model fitting of our DCC-GNGARCH is not merely under the Gaussian assumption, but persists when the innovation distribution moves to heavier tails.

Supporting Information S3: Table 12 shows the out-of-sample DM-HAC statistic comparison under the QLIKE loss, and the pattern also remains stable after replacing Gaussian innovations by Student- $t$  innovations with fixed  $\nu = 6$ . Relative to the best fit univariate GARCH(1,1) benchmark, the DCC-GNGARCH model still yields a negative DM-HAC statistic for a majority of stocks, namely, 48 and 42 out of 75 under the 30- and 60-day rolling realized variance proxies, respectively, which is broadly comparable to the Gaussian case. Against the VTE panel-GARCH(1,1) benchmark, the results remain more balanced, with 35 stocks showing negative DM-HAC statistics under both proxy windows, confirming that the panel-GARCH model remains a highly competitive benchmark even under the heavy-tailed assumption. Relative to RiskMetrics, the proposed model performs slightly less favorably under the 30-day rolling proxy, but more strongly under the 60-day rolling proxy, where 54 out of 75 stocks yield negative DM-HAC statistics. Finally, compared with Zhou's network GARCH model, the DCC-GNGARCH model still outperforms a majority of stocks, with 42 and 44 negative DM-HAC statistics under the 30- and 60-day rolling proxies, respectively, although this advantage is somewhat weaker than under Gaussian innovations. Overall, these results indicate that allowing for heavy-tailed errors does not overturn the main forecasting conclusions from the Gaussian analysis, even though the relative margins of superiority become less pronounced in some pairwise comparisons.

The evaluation of the MZ  $R^2$  regressions in Supporting Information S3: Table 13 reveals a subtle but important shift in the models' ability to track volatility trends. While the network-based models collectively continue to dominate the univariate GARCH, the panel-GARCH, and RiskMetrics baselines across both distributions, the heavy-tailed assumption distinctly enhances the trend tracking performance of Zhou's model. Specifically, under the Student- $t$  specification, Zhou's network model secures the highest MZ  $R^2$  for the largest proportion of the panel (26 and 23 stocks for the 30- and 60-day proxies, respectively), overtaking the DCC-GNGARCH model (19 and 22 stocks).

To conclude, these robustness checks suggest that the main empirical conclusions of this study remain broadly stable, but also reveal some important caveats. Across different network realized variance proxy windows, and alternative innovation distributions, the DCC-GNGARCH framework continues to show its value, especially in terms of likelihood-based fit via information criteria, competitive DM-HAC forecasting performance, and the tracking volatility directional movements via MZ  $R^2$ . At the same time, the magnitude of this advantage depends on the chosen network topology, benchmark model, and error specification. In particular, for the out-of-sample predictive accuracy, while the proposed model generally remains favorable relative

to isolated univariate benchmarks and often performs strongly against Zhou's network GARCH model, its advantage over highly competitive panel-GARCH benchmarks is more selective and is not uniformly obvious for all types of networks across the full cross-sectional stock panel. Moreover, under heavy-tailed Student- $t$  innovations, the fitted DCC-GNGARCH model exhibits its slightly weaker volatility trend-tracking performance than its network-based alternatives.

#### 4.9 | Network Sparsity and Scalability

To end this section, we carry the sparsity check as mentioned at the end of Section 4.3 to see whether the network sparsity will cause model overfitting. We re-estimate the DTF-based models under three quantile thresholds, namely, 0.02, 0.20, and 0.80, corresponding to a very sparse network (56 edges, with edge density 0.02), a moderately sparse network (555 edges, with edge density 0.20), and a highly dense network (2220 edges, with edge density 0.80). These results can be seen in Supporting Information S3: Tables 14–16.

As shown in Supporting Information S3: Table 14, when the network becomes denser, the in-sample AIC and BIC decreases. However, this trend of superiority does not reflect to corresponding out-of-sample model performance improvements, which are confirmed by the out-of-sample DM and MZ tests.

From Supporting Information S3: Table 15, the fitted DCC(1,1)-GNGARCH(1,1,[1],[1]) model appears to achieve its strongest overall out-of-sample forecasting performance at the intermediate sparsity level. For both the 30- and 60-day rolling realized variance proxies, the DTF network at quantile 0.20 yields the highest number of negative DM-HAC statistics relative to the best fit univariate GARCH(1,1), RiskMetrics, and Zhou's network GARCH benchmarks. In terms of statistically significant negative DM-HAC statistics, the quantile 0.20 specification is again strongest in most comparisons, although there are some exceptions, most notably against the panel-GARCH(1,1) benchmark and, to a minor extent, RiskMetrics, where the extremely sparse network at quantile 0.02 performs comparably or slightly better. This suggests that while a moderate network density is generally most effective for predictive accuracy, extremely sparse networks may still be advantageous when competing against strong heterogeneous benchmarks. Nevertheless, quantile 0.20 remains the preferred specification here, since it combines an economically meaningful network structure with the most consistent forecast gains across the majority of benchmarks. By contrast, the densest DTF network at quantile 0.80 produces the fewest forecasting improvements in almost all comparisons (the only exception is 30-day rolling realized variance comparison with Zhou's model, where it has 43 negative DM-HAC statistics with 28 of them being statistically significant, more than that of the sparsest DTF network), and such overall decline in number of negative DM-HAC statistic shows that increasing network density beyond a moderate level introduces unnecessary noise and weakens out-of-sample predictive performance.

Supporting Information S3: Table 16 provides further evidence on the effect of network sparsity through the out-of-sample MZ  $R^2$

results. Although the fitted DCC(1,1)-GNGARCH(1,1,[1],[1]) model achieves the highest count of best MZ  $R^2$  values among all models across all DTF networks and realized variance proxies, its strongest performance occurs at the intermediate sparsity level. In particular, that moderate DTF network with quantile 0.20 gives the largest proportion of stocks for which the proposed model delivers the highest MZ  $R^2$ , outperforming both the sparser and the denser alternatives. For the 30-day rolling realized variance proxy, both the sparsest and densest networks give the weakest result for the proposed DCC(1,1)-GNGARCH(1,1,[1],[1]) model, with the count of best MZ  $R^2$  values falling to 24. For the 60-day rolling realized variance proxy, the densest network behaves worst within the DCC-GNGARCH group, with the count dropping to 23. These findings suggest that increasing network density does not always improve, and sometimes may even weaken, the ability of network-based models to capture the overall pattern of out-of-sample volatility movements.

Together with our experiments in Section 4.8, we see that the forecasting gains of the proposed model are robust to moderate changes in network sparsity, for example, DTF network with quantile 0.10, 0.20, and 0.30, but also indicate that increasing edge density beyond a moderate level mainly improves in-sample fit rather than out-of-sample performance, hence increasing overfitting risk. To conclude, a moderately sparse network appears to provide a favorable balance between capturing cross-sectional spillovers and avoiding unnecessary network complexity.

## 5 | Discussion

This paper develops a novel network GARCH framework, the generalized network GARCH model under the dynamic conditional correlation framework (DCC-GNGARCH), which embeds GARCH dynamics within the GNAR framework so that an asset's volatility, together with its co-volatility with other assets, is influenced both by its own past and by neighboring nodes in a constructed network. The analytical contribution includes the derivation of the model, its vectorization, and connection with VARMA-type representations, together with simulation experiments based on the DCC(1,1)-GNGARCH(1,1,[1],[1]) specification, which show that the model is valid and able to reproduce key stylized facts of financial return series. We further apply the DCC(1,1)-GNGARCH(1,1,[1],[1]) model to a real-world dataset of 75 active US stocks, comparing its conditional variance and covariance forecasts for selected stocks with those of other commonly used volatility models, while also conducting a range of robustness checks to assess the stability of the main findings.

Compared with contemporary studies on network GARCH models, the DCC-GNGARCH framework offers two complementary advantages. First, it provides a parsimonious way to model multivariate volatility under a network structure, requiring substantially fewer parameters than classical multivariate ARCH-GARCH formulations such as BEKK, DCC-GARCH, or general VARMA-based specifications. Second, it accommodates higher order neighborhood effects together with dynamic covariance updates, thereby offering a mechanism through which shocks may propagate along network paths at different

distances and providing a useful tool for studying time-varying co-volatility under a network structure.

The main limitations of the present study are both methodological and empirical. From a theoretical perspective, we do not establish a general proof of stationarity for the full DCC-GNGARCH model, and the DCC correlation recursion itself is not directly network-augmented. Empirically, we focus only on the DCC(1,1)-GNGARCH(1,1,[1],[1]) specification and do not explore more general extensions involving higher order neighborhood structures, where additional regularization may be required if order is increased to result more parameters. Future work may address these gaps, extend the framework to models such as the DCC-GTN-GNGARCH, and investigate applications of network GARCH models to risk management problems such as Value-at-Risk (VaR) forecasting.

In conclusion, the proposed DCC-GNGARCH framework provides a robust and parsimonious approach to modeling high-dimensional volatility when meaningful network relations are present, while also giving analysts a useful tool for tracing how volatility spillovers propagate through a constructed network.

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### Conflicts of Interest

The author declares no conflicts of interest.

### Data Availability Statement

The raw daily stock closing price data used for computing the return series used in Section 4 can be retrieved from Yahoo Finance, accessed 19 June 2025: <https://finance.yahoo.com/markets/stocks/most-active/>. The stock universe consists of 75 selected US stocks, where related data from April 29, 2022 to December 31, 2022 were used to construct the stock network, and daily log-returns computed from closing prices over the period January 1, 2023 to July 31, 2025 were then used for model fitting, whereas data from August 1st, 2025 to February 28th, 2026 were used to construct the test log-return dataset. All code used to download, preprocess the price data, compute log-returns, estimate the models, and reproduce the results is publicly available at [https://github.com/PZhou114/DCC-GNGARCH\\_coding](https://github.com/PZhou114/DCC-GNGARCH_coding). The repository contains the processed dataset and step-by-step instructions in the README to reproduce the analysis. The processed data and scripts can also be provided by the corresponding author upon reasonable request.

### Endnotes

<sup>1</sup> To avoid confusion, from here, we replace the use of  $\Sigma_t$  by  $\hat{\Sigma}_t$  to distinguish from the true, observed conditional variance  $\Sigma_t$  and the DCC-GNGARCH model conditional variance  $\hat{\Sigma}_t$ , both in vectorized form.

<sup>2</sup> In such correlation-based constructions, potential spurious relationships should be treated with caution, since apparent return comovements may not always reflect genuine dependence between stocks. Following Granger and Newbold (1974), we compare the

coefficient of determination ( $R^2$ ) with the Durbin–Watson statistic ( $d$ ) for all  $\binom{75}{2} = 2775$  stock pairs, and no pair satisfies  $R^2 > d$ , suggesting the absence of spurious relationships in this dataset.

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## Supporting Information

Additional supporting information can be found online in the Supporting Information section. A\_new\_implementation\_of\_network\_GARCH\_Model\_for\_stock\_volatility\_and\_co\_volatility\_forecasting\_supplementary.pdf.